

PRODUCT PORTFOLIO

3 kW to 10 MW



Cleaner . Reliable . Flexible

Better power for a
limitless
Tomorrow

Cleaner . Reliable . Flexible



A RICH HERITAGE OF OVER A CENTURY OF ENGINEERING EXCELLENCE.

Kirloskar powering generating sets prioritize user experience, delivering exceptional features and benefits. Streamlined installation and enhanced dependability to expedited service, reduced maintenance costs, and optimized performance.

Kirloskar Powergen sets itself apart with groundbreaking engineering that establishes new industry benchmarks.

limitless **POTENTIAL, SUSTAINABLE PRACTICES**

Our state-of-the-art manufacturing facility embodies our commitment to sustainable practices. We partner with nature to power the facility itself, transforming waste into valuable resources. This focus on sustainability inspires both our workforce and surrounding communities. It's here, where cutting-edge technology meets exceptional skills, that we engineer solutions to empower limitless possibilities.



Discover our Plant with a
QR Code Scan.



↗ INDIA'S GENSET LEADER

KIRLOSKAR POWERGEN,
COMMITTED TO NET ZERO

\$ 1.7B

GROUP REVENUE

\$ 5.4B

MARKET CAPITALIZATION

60+

COUNTRIES



135
YEARS OF LEGACY



↗ WE'LL SECURE YOUR ENERGY NEEDS

HELPING ACHIEVE YOUR DECARBONIZATION GOALS



DECARBONIZATION

Lower Emission through
more stringed norms



RENEWABLE ENERGY

Hydrogen, Methanol,
Ethanol, Natural Gas



DISTRIBUTED ENERGY

Solar/Wind, Fuel
Agnostic Genset BESS



SMART GRIDS

AI and ML Leveraged for
Predictive Maintenance
and Grid Management

OFFERING THE BEST IN CLASS POWER SOLUTIONS!



Technology Leader

Best in class technology, proven across geographies
We will help you meet Emissions and Decarbonization goals



Largest Range

Range starting from 3kW up to 10 MW /
multiple MW utility scale solutions



Segments and Solutions Approach

End to end solution built up from power
consulting to installation & Commissioning.
Major segment includes Data Centers, Real
Estate, Metro, Rail, Defence, Pharma,
Manufacturing and Process Industries.



korloskar
powergen

Best Service Availability

3500+ technicians across the country
400 Touch points for customers



Fuel Agnostic Platforms

Diesel, Petrol, Methanol, Ethanol,
Bio-Gas, H2:Diesel



Made in India, for the World

Reach in 60 nations, 4 continents, 90% Supply
chain indigenised in India

OUR PRODUCT RANGE

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powergen



3 kW to 10 MW

We are currently executing 6.3 MW X 10 orders for a customer in India

KIRLOSKAR POWERGEN PRODUCT PORTFOLIO

3 kW - 10 MW / 12500 kVA



Kirloskar Powergen Advanced Manufactured Gensets

kVA	Fuel	Emission Norms	kVA	Fuel	Emission Norms	Model No.	kVA	Fuel	Emission Norms	Model No.
MOBILE & STATIONARY			HHP, OPTIPRIME & CUSTOM SOLUTIONS				UHHP, OPTIPRIME & CUSTOM SOLUTIONS			
3 kW	Petrol	CPCBIV+	125 kVA	Diesel	CPCBIV+	Dual Core 4R810	2500 kVA	Diesel	Stack	Dual Core DV16
4 kW	Petrol	CPCBIV+	250 kVA	Diesel	CPCBIV+	Dual Core 4K1080	2500 kVA	Diesel	Stack	Dual Core K12
5 kW	Petrol	CPCBIV+	320 kVA	Diesel	CPCBIV+	Dual Core 6K1080	2500 kVA	Diesel	CPCBIV+/Stack	Quad Core DV10
3.5 kVA	Diesel	CPCBIV+	400 kVA	Diesel	CPCBIV+	Dual Core 6K1080	3000/3300 kVA	Diesel	Stack	Dual Core K12
5 kVA	Diesel	CPCBIV+	500 kVA	Diesel	CPCBIV+	Dual Core 6K1080	3000 kVA	Diesel	CPCBIV+/Stack	Quad Core DV12
7.5kVA	Diesel	CPCBIV+	500 kVA HD	Diesel	CPCBIV+	Dual Core SL90	4000 kVA	Diesel	CPCBIV+/Stack	Octacore PV6
10 kVA	Diesel	CPCBIV+	500 kVA HD	Diesel	CPCBIV+	Quad Core 4K1080	4000 kVA	Diesel	Stack	Quad Core DV16
15 kVA	Diesel	CPCBIV+	640 kVA	Diesel	CPCBIV+	Dual Core SL90	5000 kVA	Diesel	Stack	Quad Core DV16
15 kVA	Ethanol	CPCBIV+	700 kVA	Diesel	CPCBIV+	Dual Core SL90	5000 kVA	Diesel	Stack	Quad Core K12
12.5 kVA	PNG	CPCBIV+	1000/1010 kVA	Diesel	CPCBIV+/Stack	Dual Core PV6	6000 / 6600 kVA	Diesel	Stack	Quad Core K12
15 kVA	PNG	CPCBIV+	1000/1010 kVA	Diesel	CPCBIV+/Stack	Quad Core 6K	8000 kVA	Diesel	Stack	Quad Core K12 (2026)
NATURAL GAS			1000/1010 kVA	PNG	CPCBIV+/Stack	Dual Core DV12	10000 kVA	Diesel	Stack	Quad Core K16 (2027)
25/30 kVA	PNG	CPCBIV+	1000/1010 kVA	PNG	CPCBIV+/Stack	Quad Core DV8	Powergen UHHP & Integrated Project Solutions			
62.5 kVA	PNG	CPCBIV+	2000 kVA	PNG	CPCBIV+/Stack	Quad Core DV12	2 MW / 2500 kVA	Diesel	Stack	Industrial Power Systems
82.5 kVA	PNG	CPCBIV+	1250 kVA	Diesel	CPCBIV+/Stack	Dual Core DV10	4 MW / 5000 kVA	Diesel	Stack	Industrial Power Systems
125 kVA	PNG	CPCBIV+	1250 kVA	Diesel	CPCBIV+/Stack	Quad Core SL90	6 MW / 7500 kVA	Diesel	Stack	Industrial Power Systems
250 kVA	PNG	CPCBIV+	1400 kVA	Diesel	CPCBIV+/Stack	Quad Core SL90	8 MW / 10000 kVA	Diesel	Stack	Industrial Power Systems
320 kVA	PNG	CPCBIV+	1500 kVA	Diesel	CPCBIV+/Stack	Dual Core DV12	10 MW / 12500 kVA	Diesel	Stack	Industrial Power Systems
500 kVA	PNG	CPCBIV+	2000 kVA	Diesel	CPCBIV+/Stack	Dual Core DV16				
			2000 kVA	Diesel	CPCBIV+/Stack	Quad Core PV6				

KIRLOSKAR POWERGEN PRODUCT PORTFOLIO

3 kW - 10 MW / 12500 kVA



Kirloskar Powergen Advanced Manufactured Gensets

kVA	Fuel	Emission Norms
HYBRID GENSETS		
10 kVA + 4 kW	Diesel+Solar+Battery	CPCBIV+
20 kVA + 8 kW	Diesel+Solar+Battery	CPCBIV+
30 kVA + 12 kW	Diesel+Solar+Battery	CPCBIV+
40 kVA + 16 kW	Diesel+Solar+Battery	CPCBIV+
ALTERNATE FUEL		
62.5 kVA	H2: CNG	CPCBIV+
62.5 kVA	Ethanol	CPCBIV+
62.5 kVA	H2: Diesel	CPCBIV+
125 kVA	H2: CNG	CPCBIV+

Kirloskar Powergen Gensets

kVA	Fuel	Emission Norms	kVA	Fuel	Emission Norms	kVA	Fuel	Emission Norms
LHP			MHP			HHP		
7.5 kVA	Diesel	CPCBIV+	320 kVA	Diesel	CPCBIV+	1010 kVA	Diesel	Stack
10 kVA	Diesel	CPCBIV+	320 kVA HD	Diesel	CPCBIV+	1250 kVA	Diesel	Stack
15 kVA	Diesel	CPCBIV+	400 kVA	Diesel	CPCBIV+	1250 kVA	Diesel	Stack
20 kVA	Diesel	CPCBIV+	500 kVA	Diesel	CPCBIV+	1500/1650 kVA	Diesel	Stack
25 kVA	Diesel	CPCBIV+	500 kVA HD	Diesel	CPCBIV+	2000 kVA	Diesel	Stack
30 kVA	Diesel	CPCBIV+	625 kVA	Diesel	CPCBIV+	2500 kVA	Diesel	Stack
40 kVA	Diesel	CPCBIV+	625 kVA HD	Diesel	CPCBIV+			
58.5 kVA	Diesel	CPCBIV+	750 kVA	Diesel	CPCBIV+			
82 kVA	Diesel	CPCBIV+						
125 kVA	Diesel	CPCBIV+						
160 kVA	Diesel	CPCBIV+						
200 kVA	Diesel	CPCBIV+						
200 kVA HD	Diesel	CPCBIV+						
250 kVA	Diesel	CPCBIV+						

^ Tolerances Apply: # With 0.845 Specific Gravity of diesel (5% Tolerance) || S These weight are for handling & transportation only, ±5% tolerance apply || * Efficiency of Alternator as per standards IEC 60034-1
 || ** For operation of outgoing breaker higher version of Synchronization controller is required || For Site Conditions other than standard operating conditions consult Kirloskar Oil Engines Ltd. || For Site specific layout consult Kirloskar Oil Engines Ltd to ammend the Genset alignment/configuration



INDEX

KIRLOSKAR POWERGEN ADVANCED MANUFACTURED GENSETS

SENTINEL GENSET

HYBRID GENSET

NATURAL GAS GENSET

METHANOL GENSET

OPTIPRIME

MICROGRID

KIRLOSKAR POWERGEN GENSETS

CPCB IV+ GENSET RANGING FROM 7.5 KVA TO 750 KVA

HHP GENSET

KIRLOSKAR POWERGEN GLOBAL GENSETS RANGE

kirloskar
powergen

ADVANCED MANUFACTURED GENSETS

SENTINEL

GENSETS

READY STEADY
POWER

3 kW - 7.5 kVA

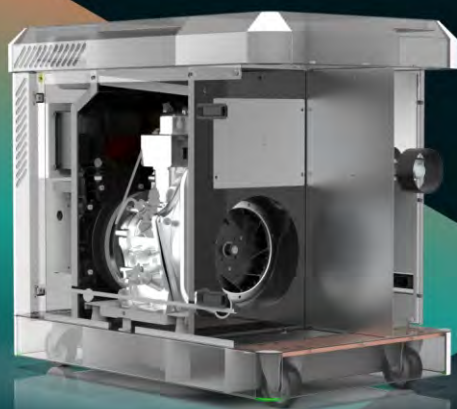


Variable Speed Diesel Generators

VSDGs are designed to deliver optimized performance, superior fuel efficiency, and lower operational costs. Unlike conventional gensets, VSDGs intelligently adjust engine speed based on load demand, ensuring minimal fuel consumption, reduced emissions, and quieter operation.

With advanced load-sensing technology, VSDGs provide cleaner power for critical load.

Ideal for applications that demand reliable, sustainable, and cost-effective power solutions.

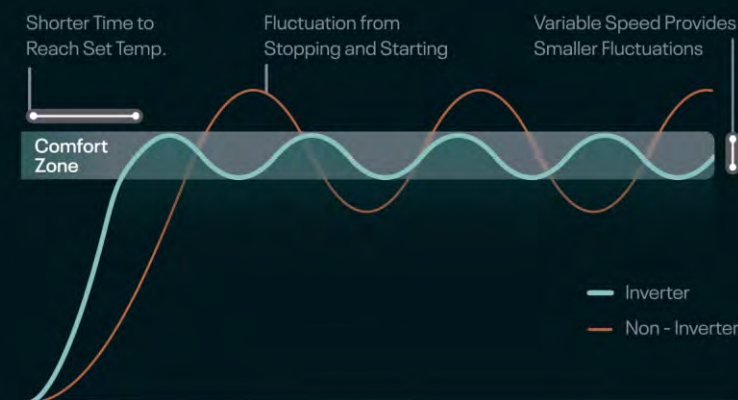


 India's #1 Genset Brand

 Inverter technology

 Easy to handle

 Low operating cost



The versatile power generator solution

Combined possible load on 3 kW*		Estimated equipment's watts	No. of units
	LED Lights	10	8
	CFL Lightning	30	8
	Tube Light	45	6
	Celling Fan	60	6
	Refrigerator	300	1

Combined possible load on 3 kW*		Estimated equipment's watts	No. of units
	Computer	300	2
	Washing machine	350	1
	TV LED Type 32"	32	2
	Air Conditioning	-	-
Any other units		-	-

* This is just for illustration & various combinations as per respective needs are possible.

** However in Case of AC operation requirement, there would be change in other connected loads and as such you are requested to contact authorised Kirloskar genset sales experts.

3 kW, 3.5 kVA , 5.5 kVA & 7.5 kVA

Petrol & Diesel Gensets Inverter Technology

kirloskar
powergen

ADVANCED MANUFACTURED GENSETS

India's #1 Genset brand

- With over 1 million Gensets in operation
- Our advanced power solutions extend up to 10 MW, ensuring reliable performance across all varying customer needs

Unmatched Service Network

- Backed by 5,000+ skilled engineers and a network of 450+ fully-equipped service centres
- Centralized monitoring ensures superior service quality and swift response times
- Hassle-free support—just contact our customer service, and an authorized representative will be at your doorstep!

Extensive Product Range

- Choose from 3 kW to 7.5 kVA gensets for optimal power solutions
- Solutions tailored for residential, and commercial applications

Advanced Power Solutions

- Aesthetically appealing enclosure with lowest footprints matching today's customer requirements
- Effortless portability for ultimate convenience — from beach parties to jungle camps
- Cutting edge VSDG technology delivering Cleaner power and lowest cost of operations
- Automatic start* for uninterrupted power
- Compliant with the latest noise and emission regulations for an eco-friendly operation



SENTINEL

3 kW - 7.5 kVA
GENSET

READY STEADY
POWER

**Why settle for limited backup?
Power a wide range of appliances
seamlessly with Kirloskar Gensets.**



Computer/
Laptop



LED
Television



Washing Machine



Air Conditioner



Refrigerator



Cold Drink
Refrigerator



Deep
Freezer

* AMF Panel is available only for diesel variants of Kirloskar Sentinel Gensets

TECHNICAL SPECIFICATIONS

Particular		PETROL			DIESEL		
Rated Output at 230V / 50Hz		3 kW	5 kW	3.5 kVA	5.5 kVA	7.5 kVA	
Max Rating***		kW/kVA	3 kW	5 kW	3.5 kVA	5.5 kVA	7.5 kVA
Genset model		KG4 - PP - 3AS	KG4 - PP - 5AS	KG4 - P - 3.5AS	KG4-P-5.5AS	KG4-P-7.5AS	
Voltage		V	230	230	230	230	
Sound Level		LWA / dB(A)	< 86 LWA			< 75 dB (A)	
Fuel Consumption**	At 100% Load		1.62	2.83	1.11	1.69	2.0
	At 75% Load	Ltrs/Hr	1.3	2.37	0.93	1.36	1.7
	At 50% Load		1	1.88	0.78	1.07	1.1
Fuel Tank Capacity		Ltrs	12.5	12.5	12.5	12.5	26
Overall Dimensions of Genset (L x W x H)^		mm	820 X 480 X 630	1040 X 550 X 685	950 X 595 X 855	950 X 595 X 855	1135 X 730 X 860
Weight of Genset with Canopy approx^		Kg	83	150	170	170	220

ENGINE

Engine Model		CCP 196	CCP389	CC 418 G1	CC 418 G1	CC 519
RPM		3000				
Rated Output (Prime Continuous rating) as per ISO 3046	kWm	4	6.5	5.5	5.5	7.2
	HP	5.5	8.8	7.47	7.47	9.8
Cubic Capacity	cc	196	389	418	418	519
Lube Oil Change Period \$	Hrs	First 20 hr. then Every 100 hr.			100	
Lube Oil Sump Capacity	Ltrs	0.6	1.1	1.65	1.65	1.7
Starting System		Electric / Recoil		Electric	Electric	Electric
Low Oil Level alert (Trip)		Yes	Yes	Low Lube Oil Level-No, Low Lube Oil Press-Yes		Low Lube Oil Press-Yes
Overload Warning		Yes	-	Yes	Yes	Yes
Battery Charging		Yes	Yes	Yes	Yes	12V, 8A SMPS Charger
Eco Throttle		Electronic	Mechanical	Mechanical	Mechanical	Mechanical

INVERTER

Rated Output at 230V / 50Hz	kW	3	3.5	6	6
Max Output	kW	3	4	6.5	6.6
% THD (Voltage)		< 5%	< 5%	< 5%	< 5%
Display Parameters		Digital Multi Fu'n Meter Voltage, Frequency, Hours		KG645-Controller (Voltage, Frequency, Hours)	Microprocessor based Genset controller
Fuel Gauge (Mech. Level Indicator)		Yes	Yes	Yes	Yes
AC Output connector		Yes	Yes	Yes	Yes
AC Circuit Breaker		Yes	Yes	Yes	Yes

Notes

^ Standard Tolerances Apply | ** 5% Tolerance ; 5% Tolerance with ECO Mode for 2.8 kW Petrol | *** Some Electrical Appliance motor take more power during startup than rated power \$ At NTP Genset Model will be available in 4 wheel configuration. For regular Operation do not exceed rated power

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ADVANCED MANUFACTURED GENSETS

HYBRID

GENSETS

10 kVA + 4 kW
to
40 kVA + 16 kW



OPTIPRIME HYBRID GENSETS

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ADVANCED MANUFACTURED GENSETS



THE VERSATILE HYBRID POWER GENSET SOLUTION



ATM



Laboratory



Telecom



House

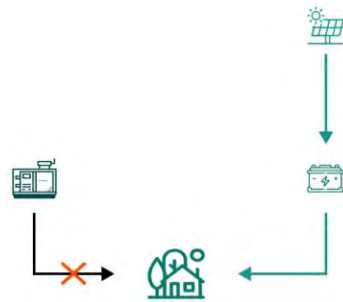


Shops

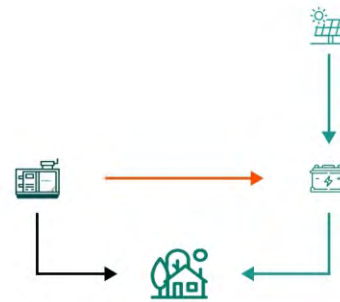
OPTIPRIME HYBRID GENSETS

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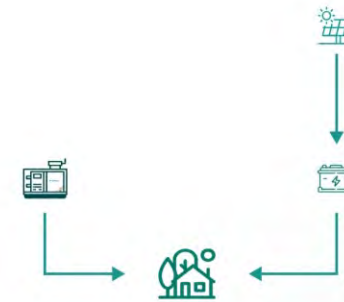
ADVANCED MANUFACTURED GENSETS



Low



Medium



High

Load

Power Required

Upto 4kW

Upto 8kW

Upto
10+4kW*

Load Fulfilled By

Battery Grid

DG + Battery Charging

DG + Battery Grid

Hybrid Solution Equipped with 10 kVA Diesel Genset can server loads up to 14kW

*For Limited Time

- Whenever SOC goes to 30%, DG will automatically start to charge the battery.
- Battery can charge through EB (Electricity Board) as well as Solar Panel (Optional)



OPTIPRIME HYBRID GENSETS

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ADVANCED MANUFACTURED GENSETS

Integrates multiple Energy Sources

Compatible with traditional fuel generators such as diesel, battery, and renewable sources like solar power

Fuel efficiency and cost reduction

Minimizes diesel engine runtime, optimized fuel consumption and hassle-free maintenance

Sustainable Green Technology

Reduces carbon dioxide emissions, and operates with ultra-low noise levels

Compact Footprint & Easy Transportability

Optimized footprint and effortless transportation, even in constrained or remote environments

Optimized Operation & Maintenance

Extended service intervals and reduced maintenance downtime for enhanced reliability

FEATURES



TECHNICAL SPECIFICATIONS

GENSET

Prime Rating at Rated RPM [as per ISO8528] ¹	kVA	10
	kW	8
Maximum Rating for 1 Hour (DG+Battery)	kVA	14
Voltage / Frequency / Power Factor	-	230 / 50 / 0.8
Noise Level	dBA	Max 70 @ 7 Mtr.
Fuel Consumption [#]	At 100% Load	2.75
	At 75% Load	2.06
	At 50% Load	1.60
	At 25% Load	1.15
Fuel Tank Capacity [Standard DG Set]	Ltrs	40
Weight of Genset with Canopy [Approx.] ^{\$}	Dry	950
	Wet	1000
Overall Dimensions of Genset [^]	Length	2300
	Width	700
	Height (Inside Silencer)	1050
	With Solar Panel Up	1180
	With Solar Panel Down	1490

ENGINE

Engine Model	-	2R550 NA
Rated Output [Prime Continuous Rating as per ISO 8528-1]	kW	10.3
	HP	14
No. of Cylinders	No.	2
Cubic Capacity ²	Ltr.	1.09
Lube Oil change period	Hrs.	500
Coolant Capacity	Ltr.	4
Block Loading Capacity	%	100%

ALTERNATOR

Insulation Class	-	H
Alternator Efficiency [at 100% Load] 0.8 pf [*]	%	83.8
Max Voltage Dip at Full Load 0.8 pf Lag	No.	2
Max time to build up Rated Voltage at Rated RPM	Ltr.	1.09

[^] Tolerances Apply

[#] With 0.845 Specific Gravity of diesel (10% Tolerance)

^{\$} These weight are for handling & transportation only, +5% tolerance apply

^{*} Efficiency of Alternator as per standards IEC 60034-1

For Site Conditions other than standard operating conditions consult Kirloskar Oil Engines Ltd.

INVERTER

Capacity	kVA	4
Battery Capacity	Ah	200
Battery System Voltage	V	51.2
Battery Type	-	LFP
Battery Management and Protection	-	Yes

PANEL

Controller Make	DSE
Operating Voltage	12

SOLAR SYSTEM -OPTIONAL

Capacity	W	160-300
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ADVANCED MANUFACTURED GENSETS

NATURAL GAS

GENSETS

15 kVA - 500 kVA



BENEFITS

READY TO USE GENSET - PLUG & PLAY

LOWER OPERATING COST

CONSISTENT FUEL QUALITY

ELECTRONIC ECU & CARBURATION SYSTEM



THE VERSATILE POWER
GENERATOR SOLUTION



FOOD INDUSTRY



HEALTHCARE



HOSPITALITY



MANUFACTURING



REAL
ESTATE

CPCB IV+

Natural Gas Gensets
(15 kVA - 500 kVA)

korloskar
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ADVANCED MANUFACTURED GENSETS

Exhaust System

- In built catalytic convertor as standard scope
- Residential silencer with Protection cover
- Latest CPCB compliant exhaust System

Controller

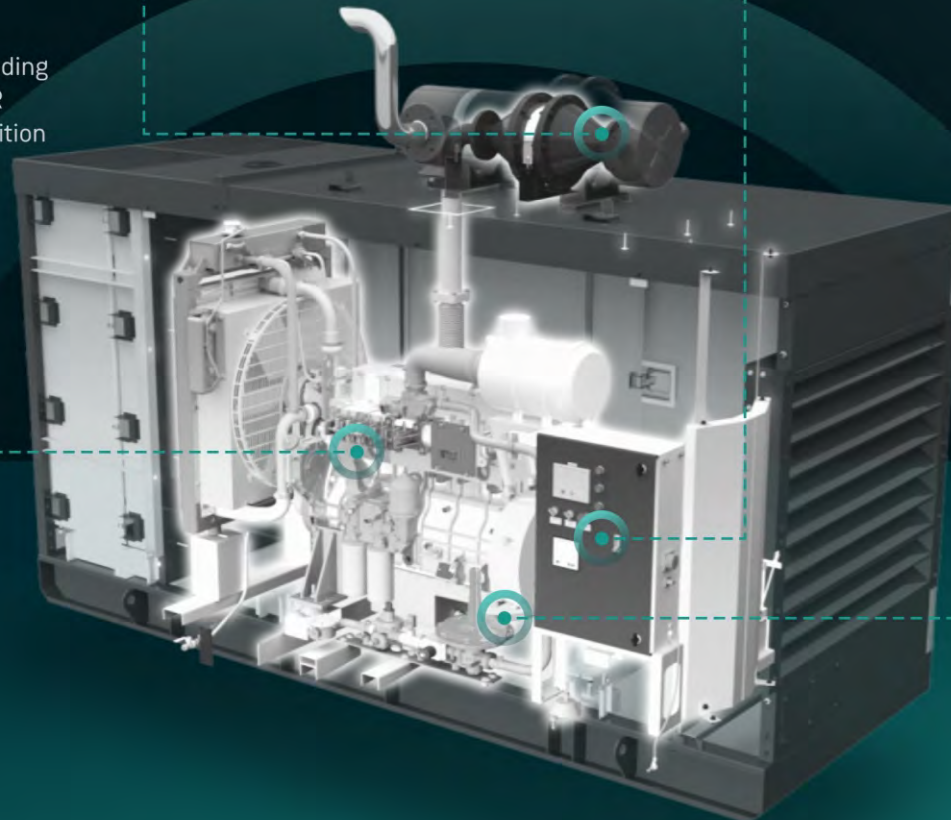
- Microprocessor based KG 645 controller
- Graphical LCD display
- Best in class Monitoring & diagnostic capability
- Integrable with AMF & Communication compatible

Engine

- Fully Electronic controlled Engine
- Higher Cubic capacity results higher block loading
- Electronic carburation system for correct AFR
- Efficient HT Coil & Spark plugs for precise ignition
- Best in class fuel efficiency

Gas Train

- CPCB approved Gas train inside acoustic enclosure
- Consist of Gas filter, Solenoid & PRVs
- Designed & developed to suit for genset application



TECHNICAL SPECIFICATIONS

Prime Rating at rated rpm (as per ISO 8528)	kVA	15 (ESP)	30	58.5	125	250	500
	kW	12	24	46.8	100	200	400
Genset Model		KG4-15AS NG1	KG4-30WS NG1	KG4-58.5WS NG1	KG4-125WS NG1	KG4-250WS NG1	KG4-500WS NG1
Frequency	Hz	50					
Power factor	lagging	0.8					
Voltage	V	230 (1Ø) & 415 (3Ø)			415 (3Ø)		
Steady-state Governing class (As per ISO 8528-5)		G3					
Noise level @ 75% rated load	dBA	< 75 @ 1 meter					
Fuel consumption* At 100% 75% 50%	SCM / Hr	4 3.5 3.1	9.1 7.1 5.5	15.5 12.5 9.9	29.3 23.2 17.8	58.8 47.2 35.7	72.2 58.5 47.2
Inlet Gas pressure requirement	Min (Bar)	1	1	0.5	0.5	1	1
	Max (Bar)	4	4	4	4	4	4
Weight of Genset with Canopy approx^ Dry	Kg	850	1100	2500	3600	5700	7800
Overall dimensions of genset L x W x H	mm	1740 x 1050 x 1410	2550 x 1122 x 1410	3325 x 1225 x 1800	3750 x 1500 x 1980	5100 x 2000 x 2410	6518 x 2123 x 2708
Electrical battery starting voltage	Volts-DC	12	12	12	24	24	24

ENGINE

Engine Model		HA294 NG1	3R1040 NG1	6R1080 NG1	SL90TA NG1	DV8TA NG1	DV12TA NG1
Rated output	kW	15.1	30.1	54.4	114.7	228	447.2
	HP	20.5	42	74	156	310	608
No. of cylinder	Number	2	3	6	6	8	12
Cubic capacity	Ltrs.	1.89	3.1	6.49	8.86	15.92	23.89
Bore x Stroke	mm	100 x 120	105X20	105 x 125	118x 135	130x150	130x150
Rated speed	rpm						
Aspiration	NA/TC/TA	NA	NA	NA	TA	TA	TA
Lube Oil change period	hrs.						
Lube oil sump capacity	Ltrs.	5	9	17	27	46	78
Coolant capacity	Ltrs.	NA	14	25	30	30	145

ALTERNATOR

Insulation Class		Class H					
Alternator efficiency (at 100% load) 0.8 pf	%	86	82.2	91	92.5	92.5	94.6
Max voltage dip at full load 0.8 pf lag		< 20 %					
Max time to build up rated voltage at rated RPM		< 5 sec provided engine reach the rated speed					

- The maximum power of engines is declared at NTP conditions and an appropriate deration factor will be applicable with respect to Natural Gas genset site altitude and ambient air conditions.
- The effect of the fuel quality should also take into account on performance, i.e. Methane %: Min 88% & Calorific Value of Fuel: Min 8500 Kcal/m3.
- A tolerance of +/- 5% will applicable as per IS 8528 Standard. Dimensions are approximate and vary with options. ^ Genset weight tolerance 50 Kg.
- SB- Standby SCM- Standard Cubic Meter Minimum 70 methane number required Above specifications are subject to change without prior notice due to continuous technical development.

karloskar
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ADVANCED MANUFACTURED GENSETS

METHANOL

GENSETS

20 kVA | 62.5 kVA



METHANOL GENSETS

Experience a power solution that is
both environmentally conscious and
operationally superior



FUEL AGNOSTIC SOLUTIONS FOR EVOLVING POWER DEMANDS

In the global pursuit of net-zero emissions, businesses are increasingly seeking environmentally responsible fuel solutions. Kirloskar is at the forefront of this transition, providing adaptable power generation systems that meet the dynamic needs of modern industries.

Our M100 methanol gensets represent a robust and reliable power solution, engineered for applications requiring customizable power generation. Utilizing methanol as a sustainable and clean fuel, these gensets achieve a significant 20% reduction in CO₂ emissions compared to conventional diesel systems.

Equipped with advanced electronic governing systems, our methanol gensets offer seamless synchronization capabilities. This feature allows for effortless integration into complex power systems, ensuring operational continuity and efficiency.

Methanol's inherent versatility allows for blending with other fuels, enabling the creation of tailored power generation solutions. Designed for quiet operation and exceptional reliability, our gensets deliver clean energy without compromising performance.

BENEFITS



100%
Methanol



Reduces CO₂
emissions by 20%



Compact and lightweight
design



Spark ignition engine
technology



ECU-controlled governing
system



High-load acceptance
compliant with ISO 8528



Environment
friendly



Remote monitoring and
Onboard diagnostics



Corrosion-resistant fuel
tank and submersible fuel
pump



Optimized Noise Vibration
and Harshness (NVH)
levels

TECHNICAL SPECIFICATIONS

GENSET

Genset Rating	20 kVA	62.5 kVA
Engine Model	2R1040NA	6R1080NA
Type	Silent	Silent
Genset Power output (kVA)	20	62.5
Genset Power output (kWe)	16	50
Frequency (Hz)	50 Hz	50 Hz
Voltage (L-L) (V)	415	415
Voltage (L-N) (V)	230	230
Power Factor	0.8	0.8
Rated Alternator Efficiency @ 0.8	86.8	91.1
Phase	1ph / 3ph	3ph
Method of Starting	Electric (12VDC)	Electric (12VDC)
Battery Capacity	75 AH	130 AH
Governing class	G2	G2
Dimensions (L x W x H) Silent Height considered without silencer (mm x mm x mm)	2085 × 960 × 1300	3326 × 1337 × 1955

ENGINE

Engine Model	2R1040NA	6R1080NA
No. of Cylinders	2	6
Configuration	Inline Water Cooled	Inline Water Cooled
Aspiration	Naturally Aspirated	Naturally Aspirated
Bore (mm)	105	105
Stroke (mm)	120	125
Engine Cubic Capacity (cc)	2078	6494
Compression Ratio	13:1	13.5:1
Valve Per Cylinder (Inlet / Exhsust)	1/1	1/1
Firing Order	1-2	1-5-3-6-2-4
Fuel	Methanol (M100)	Methanol (M100)
Combustion	Stoichiometric, Premixed and spark ignited	
Rated Power (HP)	25.5	83
Rated Power (kW)	18.75	61
Rated RPM	1500	1500
Rated BMEP (Bar)	7.36	7.7

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ADVANCED MANUFACTURED GENSETS

OPTIPRIME

125 kVA - 6600 kVA



OPTIPRIME

Advanced Manufactured Gensets

125 kVA - 6600 kVA



Multi-Core Power Systems

The OPTIPRIME Multi-Core (Dual-Core, Quad-Core, Octa-Core) power systems are custom-engineered solutions tailored to meet the specific needs of each customer's site. These turnkey solutions provide a compact design with exceptional noise reduction. Powered by patented OPTIPRIME technology, they deliver remarkable improvements in total cost of ownership and industry-leading fuel efficiency. The multicore solution offers breakthrough redundancy, further enhancing reliability and ensuring continuous power. Supported by Kirloskar's warranty, customers benefit from reduced operational costs, greater savings, and a minimized footprint. Its versatile design allows for single-unit deployment or combined prime and standby functionality within a single enclosure, providing optimal performance and adaptability for varying power needs.

The Versatile Power Solution



Infrastructure



Industry



IT/TES



Hospitality



Health



Realty



Government
& Defense



Logistics

OPTIPRIME

kirloskar
powergen

ADVANCED MANUFACTURED GENSETS

BENEFITS

Kirloskar's OPTIPRIME delivers a customized and efficient power solution for applications with fluctuating loads. By intelligently managing multi-core operations through a centralized control panel and integrated Power Management System, OPTIPRIME dynamically adjusts to real-time load demands, significantly reducing fuel consumption.



Patented
Hybrid
Technology



Multi-core operations
with In-built
Synchronisation



Reduced Carbon
Dioxide Emissions
by 40%



Optimised Fuel
Consumption



Reduced
Ownership Cost



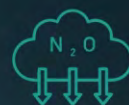
Optimum
Power



Optimal
Efficiency



Better
Flexibility



Reduced Nitrous
Oxide Emissions
by 50%

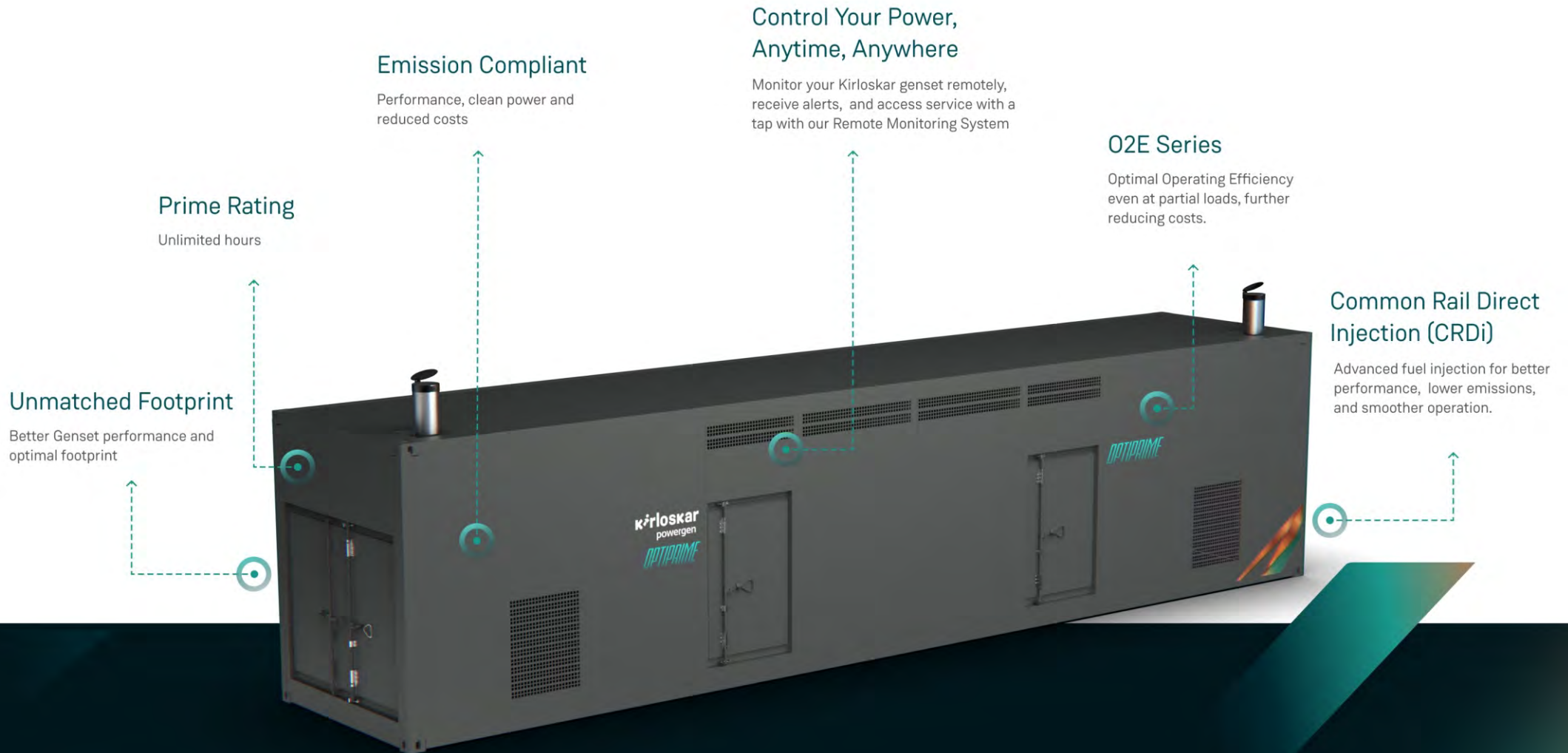


Lower Product
Footprint -20%
Reduction in Space

OPTIPRIME FEATURES

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ADVANCED MANUFACTURED GENSETS





125 kVA to 6600 kVA Largest Portfolio in the World !

kVA	Configuration Options*	Fuel	Emission Norms	Engine Configurations
125 kVA	3	Diesel	CPCBIV+	Dual Core 4R810
250 kVA	3	Diesel	CPCBIV+	Dual Core 4K1080
320 kVA	4	Diesel	CPCBIV+	Dual Core 6K1080
400 kVA	3	Diesel	CPCBIV+	Dual Core 6K1080
500 kVA	2	Diesel	CPCBIV+	Dual Core 6K1080
500 kVA HD	2	Diesel	CPCBIV+	Dual Core SL90
500 kVA HD	2	Diesel	CPCBIV+	Quad Core 4K1080
640 kVA	3	Diesel	CPCBIV+	Dual Core SL90
700 kVA	2	Diesel	CPCBIV+	Dual Core SL90
1000/1010 kVA	1	Diesel	CPCBIV+/Stack	Dual Core PV6
1000/1010 kVA	2	Diesel	CPCBIV+/Stack	Quad Core 6K
1000/1010 kVA	3	PNG	CPCBIV+/Stack	Dual Core DV12
1000/1010 kVA	3	PNG	CPCBIV+/Stack	Quad Core DV8
2000 kVA	3	PNG	CPCBIV+/Stack	Quad Core DV12
1250 kVA	3	Diesel	CPCBIV+/Stack	Dual Core DV10
1250 kVA	2	Diesel	CPCBIV+/Stack	Quad Core SL90

kVA	Configuration Options*	Fuel	Emission Norms	Engine Configurations
1400 kVA	2	Diesel	CPCBIV+/Stack	Quad Core SL90
1500 kVA	4	Diesel	CPCBIV+/Stack	Dual Core DV12
2000 kVA	4	Diesel	CPCBIV+/Stack	Dual Core DV16
2000 kVA	1	Diesel	CPCBIV+/Stack	Quad Core PV6
2250 kVA	3	Diesel	Stack	Dual Core DV16
2500 kVA	4	Diesel	Stack	Dual Core DV16
2500 kVA	4	Diesel	Stack	Dual Core K12
2500 kVA	4	Diesel	CPCBIV+/Stack	Quad Core DV10
3000/3300 kVA	4	Diesel	Stack	Dual Core K12
3000 kVA	4	Diesel	CPCBIV+/Stack	Quad Core DV12
4000 kVA	3	Diesel	CPCBIV+/Stack	Octacore PV6
4000 kVA	4	Diesel	Stack	Quad Core DV16
5000 kVA	4	Diesel	Stack	Quad Core DV16
5000 kVA	4	Diesel	Stack	Quad Core K12
6000/6600 kVA	3	Diesel	Stack	Quad Core K12
8000 kVA	3	Diesel	Stack	Quad Core K12 (2026)
10000 kVA	3	Diesel	Stack	Quad Core K16 (2027)

*Configuration options available: Series/ Parallel/ Stacking etc.

Powergen HHP & Integrated Project Solutions

2 MW / 2500 kVA	Diesel	Stack	Industrial Power Systems
4 MW / 5000 kVA	Diesel	Stack	Industrial Power Systems
6 MW / 7500 kVA	Diesel	Stack	Industrial Power Systems
8 MW / 10000 kVA	Diesel	Stack	Industrial Power Systems
10 MW / 12500 kVA	Diesel	Stack	Industrial Power Systems



GENSET SPECIFICATIONS

125 - 700 kVA

		OPTIPRIME 125	OPTIPRIME 250	OPTIPRIME 320	OPTIPRIME 400	OPTIPRIME 500	OPTIPRIME 500 HD	OPTIPRIME 500 HD	OPTIPRIME 640	OPTIPRIME 700
		DUAL CORE 4R810	DUAL CORE 4K1080	DUAL CORE 6K1080	DUAL CORE 6K1080	DUAL CORE 6K1080	DUAL CORE SL90	Quad Core 4K1080	DUAL CORE SL90	DUAL CORE SL90
Rating at Rated RPM	kVA	125	250	320	400	500	500	500	640	700
	kW	100	200	256	320	400	400	400	512	560
Voltage / Frequency / Power Factor		415V / 50HZ / 0.8 lagging								
Noise Level	dBA	≤ 75 dBA @ 1Mtr (Genset with canopy)								
Configuration Options		3	3	4	3	2	2	3	3	2
Fuel		Diesel	Diesel	Diesel	Diesel	Diesel	Diesel	Diesel	Diesel	Diesel
Fuel Consumption*	At 100% Load	21.6	48	74	91.4	106	89	97	133	89
	At 75% Load	16.6	37	58	71.1	80	69	76	102	69
	At 50% Load	12.6	26	41	50.2	58	49	54	69	49
	At 25% Load	8.9	11	16	25.1	30	27	32	35	27
Fuel Tank Capacity	Ltrs.	165 Dual Core	230 Dual Core	400 Dual Core	400 Dual Core	400 Dual Core	500 Dual Core	230 Quad Core	500 Dual Core	500 Dual Core
Tentative dimensions *Customized solution, actual may vary as per customer requirement and layout	Length	mm	2900	2520	2993	7500	3050	8500	5030	8500
	Width	mm	2100	2770	2130	1450	2130	1700	2770	1700
	Height (With Silencer)	mm	1701	1570	1622	1900	1422	2042	1572	2042
	Genset Alignment / Configuration		Parallel	Parallel	Parallel	Series	Parallel	Series	Parallel	Series
Dry weight of Genset with Canopy (Approx.)		Kg	2511	5022	5960	6000	6160	8000	7900	8100

^ Tolerances Apply: # With 0.845 Specific Gravity of diesel (5% Tolerance) || \$ These weight are for handling & transportation only, ±5% tolerance apply || * Efficiency of Alternator as per standards IEC 60034-1 || ** For operation of outgoing breaker higher version of Synchronization controller is required || For Site Conditions other than standard operating conditions consult Kirloskar Oil Engines Ltd. || For Site specific layout consult Kirloskar Oil Engines Ltd to ammend the Genset alignment/configuration

*Product specifications & dimensions are subject to change without notice.

ENGINE SPECIFICATIONS

ENGINE MODEL		OPTIPRIME 125 4R810ETA 4G1	OPTIPRIME 250 4K1080ETA 4G1	OPTIPRIME 320 6K1080ETA 4G1	OPTIPRIME 400 6K1080ETA 4G2	OPTIPRIME 500 6K1080ETA 4G2	OPTIPRIME 500 DUAL CORE SL90	OPTIPRIME 500HD 4K1080ETA 4G1	OPTIPRIME 640 DUAL CORE SL90	OPTIPRIME 700 DUAL CORE SL90
Rated output (Prime continuous ratings as per ISO 8528-1)	kW	54.4 Dual Core	114.7 Dual Core	147.1 Dual Core	183.8 Dual Core	183 Dual Core	228 Dual Core	114.7 Quad Core	279.5 Dual Core	279.5 Dual Core
	HP	74 Dual Core	156 Dual Core	200 Dual Core	250 Dual Core	250 Dual Core	310 Dual Core	156 Quad Core	380 Dual Core	380 Dual Core
Number of Cylinders	No.	4 Dual Core	4 Dual Core	6 Dual Core	6 Dual Core	6 Dual Core	6 Dual Core	4 Quad Core	6 Dual Core	6 Dual Core
Cubic Capacity	Ltrs.	3.24 Dual Core	4.32 Dual Core	6.48 Dual Core	6.48 Dual Core	6.64 Dual Core	8.86 Dual Core	4.32 Quad Core	8.86 Dual Core	8.86 Dual Core
Lube Oil Change Period	Hrs.	500	500	500	500	500	500	500	500	500
Lube Oil Sump Capacity	Ltrs.	10 Dual Core	14 Dual Core	25 Dual Core	25 Dual Core	25 Dual Core	27 Dual Core	14 Quad Core	27 Dual Core	27 Dual Core
Coolant Capacity	Ltrs.	12.70 Dual Core	17.7 Dual Core	28.9 Dual Core	28.9 Dual Core	28.9 Dual Core	36.4 Dual Core	17.7 Quad Core	36.4 Dual Core	36.4 Dual Core
AdBlue / DEF Capacity	Ltrs.	NA	25 Dual Core	25 Dual Core	45 Dual Core	45 Dual Core	45 Dual Core	25 Quad Core	45 Dual Core	45 Dual Core

ALTERNATOR SPECIFICATIONS

Insulation Class		Class H	Class H	Class H	Class H	Class H	Class H	Class H	Class H	Class H
Max Voltage Dip at Full Load 0.8 pf Lag	%	<20%	<20%	<20%	<20%	<20%	<20%	<20%	<20%	<20%
Max Time to build up rated voltage at Rated RPM	Sec	< 2 sec provided engine reach the rated speed								

MULTI-CORE POWER SYSTEMS SUITABLE FOR HT ALTERNATOR

*Product specifications & dimensions are subject to change without notice.

** For operation of outgoing breaker higher version of Synchronization controller is required

GENSET SPECIFICATIONS

1000 - 6600 kVA

		OPTIPRIME 1000 <i>World smallest footprint</i>	OPTIPRIME 1000 QUAD CORE 6K1080	OPTIPRIME 1000 DUAL CORE DV12	OPTIPRIME 1000 QUAD CORE DV8	OPTIPRIME 1250 DUAL CORE DV10	OPTIPRIME 1250 QUAD CORE SL90	OPTIPRIME 1400 QUAD CORE SL90	OPTIPRIME 1500 DUAL CORE DV12
Rating at Rated RPM	kVA	1000	1000	1000	1000	1250	1250	1400	1500
	kW	800	800	800	800	1000	1000	1120	1200
Voltage / Frequency / Power Factor		415V/50Hz/0.8Pf	415V/50Hz/0.8Pf	415V/50Hz/0.8Pf	415V/50Hz/0.8Pf	415V/50Hz/0.8Pf	415V/50Hz/0.8Pf	415V/50Hz/0.8Pf	415V/50Hz/0.8Pf
Noise Level		75dBA at 1 m	75dBA at 1 m	75dBA at 1 m	75dBA at 1 m	75dBA at 1 m	75dBA at 1 m	75dBA at 1 m	75dBA at 1 m
Configuration Options		1	2	3	3	3	2	2	4
Fuel		Diesel	Diesel	PNG	PNG	Diesel	Diesel	Diesel	Diesel
Fuel Consumption*	At 100% Load	194	178	180 Kg/hr	177.8 Kg/hr	248	212	264	292
	At 75% Load	152	138	146 Kg/hr	143.5 Kg/hr	187	160	204	240
	At 50% Load	123	98	118 Kg/hr	108.5 Kg/hr	126	116	138	170
Fuel Tank Capacity		990	400 Quad Core	Not Applicable	Not Applicable	990 Dual Core	600 Quad Core	600 Quad Core	990 Quad Core
Tentative dimensions *Customized solution, actual may vary as per customer requirement and layout	Length	3400	4870	3362	3410	2960	5094	5094	3803
	Width	2400	2130	3170	4025	3170	5108	2554	3170
	Height (With Silencer)	2900	2220	1997	2610	1982	4476	1772	2125
	Genset Alignment / Configuration	Kg Parallel	Stacking	Parallel	Parallel	Parallel	Stacking	Parallel	Parallel
Dry weight of Genset with Canopy (Approx.)		10800	13786	17846	22800	16200	15960	17500	18500

^ Tolerances Apply: # With 0.845 Specific Gravity of diesel (5% Tolerance) || \$ These weight are for handling & transportation only, ±5% tolerance apply || * Efficiency of Alternator as per standards IEC 60034-1 || ** For operation of outgoing breaker higher version of Synchronization controller is required || For Site Conditions other than standard operating conditions consult Kirloskar Oil Engines Ltd. || For Site specific layout consult Kirloskar Oil Engines Ltd to ammend the Genset alignment/configuration

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ENGINE SPECIFICATIONS

		OPTIPRIME 1000	OPTIPRIME 1000	OPTIPRIME 1000	OPTIPRIME 1000	OPTIPRIME 1250	OPTIPRIME 1250	OPTIPRIME 1400	OPTIPRIME 1500
		World smallest footprint							
Engine Model		D15L6	6K1080ETA 4G2	DV12TA NG1	DV8TA NG1*	DV10ETA 4G2	6SL90ETA 4G3	6SL90ETA 4G3	DV12ETA 4G2
Rated output (Prime continuous ratings as per ISO 8528-1)	kW	447 Dual Core	183.8 Quad Core	447.2 Dual Core	228 Quad Core	561.1 Dual Core	279.5 Quad Core	279.5 Quad Core	662 Dual Core
	HP	608 Dual Core	250 Quad Core	608 Dual Core	310 Quad Core	763 Dual Core	380 Quad Core	380 Quad Core	900.6 Dual Core
Number of Cylinders	No.	6 Dual Core	6 Quad Core	12 Dual Core	8 Quad Core	10 Dual Core	6 Quad Core	6 Quad Core	12 Dual Core
Cubic Capacity	Ltrs	14.76 2	6.48 Quad Core	23.89 Dual Core	15.92 Quad Core	19.9 Dual Core	8.86 Quad Core	8.86 Quad Core	23.88 Dual Core
Lube Oil Change Period	hrs.	500	500	500	500	500	500	500	500
Lube Oil Sump Capacity	Ltrs	42 Dual Core	25 Quad Core	78 Dual Core	46 Quad Core	50 Dual Core	27 Quad Core	27 Quad Core	73 Dual Core
Coolant Capacity	Ltrs	250 Dual Core	28.9 Quad Core	145 Dual Core	30 Quad Core	81.7 Dual Core	36 Quad Core	36 Quad Core	173.9 Dual Core
AdBlue / DEF Capacity	Ltrs	45 Dual Core	45 Quad Core	NA	NA	45 Dual Core	45 Quad Core	45 Quad Core	45 Dual Core

ALTERNATOR SPECIFICATIONS

Insulation Class		Class H	Class H	Class H	Class H	Class H	Class H	Class H	Class H
Max Voltage Dip at Full Load 0.8 pf Lag	%	<20%	<20%	<20%	<20%	<20%	<20%	<20%	<20%
Max Time to build up rated voltage at Rated RPM	Sec	< 2 sec provided engine reach the rated speed							

MULTI-CORE POWER SYSTEMS SUITABLE FOR HT ALTERNATOR

** For operation of outgoing breaker higher version of Synchronization controller is required

*Product specifications & dimensions are subject to change without notice.

GENSET SPECIFICATIONS

1000 - 6600 kVA

		OPTIPRIME 2000 DUAL CORE DV16	OPTIPRIME 2000 QUAD CORE PV6	OPTIPRIME 2000 QUAD CORE DV12	OPTIPRIME 2250 DUAL CORE DV16	OPTIPRIME 2500 DUAL CORE DV16	OPTIPRIME 2500 DUAL CORE K12	OPTIPRIME 2500 QUAD CORE DV10
Rating at Rated RPM	kVA	2000	2000	2000	2250	2500	2500	2500
	kW	1600	1600	1600	1800	2000	2000	2000
Voltage / Frequency / Power Factor		415V/50Hz/0.8Pf	415V/50Hz/0.8Pf	415V/50Hz/0.8Pf	415V/50Hz/0.8Pf	415V/50Hz/0.8Pf	415V/50Hz/0.8Pf	415V/50Hz/0.8Pf
Noise Level		75dBA at 1 m	75dBA at 1 m	75dBA at 1 m	75dBA at 1 m	75dBA at 1 m	75dBA at 1 m	75dBA at 1 m
Configuration Options		1	1	3	3	4	4	4
Fuel		Diesel	Diesel	PNG	Diesel	Diesel	Diesel	Diesel
Fuel Consumption*	At 100% Load	380	388	360 Kg/hr	380	380	475	496
	At 75% Load	294	302	292 Kg/hr	294	294	359	374
	At 50% Load	213	246	236 Kg/hr	213	213	252	252
Fuel Tank Capacity		990 Quad Core	990 Quad Core	Not Applicable	990 Dual Core	990 Dual Core	External Tank	990 Quad Core
Tentative dimensions *Customized solution, actual may vary as per customer requirement and layout	Length	7800	6800	3362	4505	5305	4603	4130
	Width	5600	2400	3170	3390	2430	4244	3170
	Height (With Silencer)	2713	2900	2554	2410	5600	2819	2154
	Genset Alignment / Configuration	Kg Parallel	Parallel	Parallel	Parallel	Stacking	Parallel	Parallel
Dry weight of Genset with Canopy (Approx.)		25170	21600	35690	26752	26400	36500	32400

^ Tolerances Apply: # With 0.845 Specific Gravity of diesel (5% Tolerance) || \$ These weight are for handling & transportation only, ±5% tolerance apply || * Efficiency of Alternator as per standards IEC 60034-1 || ** For operation of outgoing breaker higher version of Synchronization controller is required

|| For Site Conditions other than standard operating conditions consult Killoskar Oil Engines Ltd. || For Site specific layout consult Killoskar Oil Engines Ltd to ammend the Genset alignment/configuration

*Product specifications & dimensions are subject to change without notice.

ENGINE SPECIFICATIONS

		OPTIPRIME 2000	OPTIPRIME 2000	OPTIPRIME 2000	OPTIPRIME 2250	OPTIPRIME 2500	OPTIPRIME 2500	OPTIPRIME 2500
Engine Model		DV16ETA G3	D15L6	DV12TA NG1	DV16ETA G3	DV16ETA G3	12K4300-E1	DV10ETA 4G2
Rated output (Prime continuous ratings as per ISO 8528-1)	kW	889 Dual Core	447 Quad Core	447 Quad Core	889 Dual Core	889 Dual Core	1097 Dual Core	561.1 Quad Core
	HP	1210 Dual Core	608 Quad Core	608 Quad Core	1210 Dual Core	1210 Dual Core	1492 Dual Core	561.1 Quad Core
Number of Cylinders	No.	16 Dual Core	6 Quad Core	6 Quad Core	16 Dual Core	16 Dual Core	12 Dual Core	10 Quad Core
Cubic Capacity	Ltrs	31.84 Dual Core	14.76 Quad Core	23.89 Quad Core	31.84 Dual Core	31.84 Dual Core	51.72 Dual Core	19.9 Quad Core
Lube Oil Change Period	hrs.	500	500	500	500	500	500	500
Lube Oil Sump Capacity	Ltrs	130 Dual Core	42 Quad Core	78 Quad Core	130 Dual Core	130 Dual Core	285 Dual Core	50 Quad Core
Coolant Capacity	Ltrs	180 Dual Core	250 Quad Core	145 Dual Core	180 Dual Core	180 Dual Core	355 Dual Core	81.7 Quad Core
AdBlue / DEF Capacity	Ltrs	NA	45 Quad Core	NA	NA	NA	NA	45 Quad Core

ALTERNATOR SPECIFICATIONS

Insulation Class		Class H	Class H	Class H	Class H	Class H	Class H	Class H
Max Voltage Dip at Full Load 0.8 pf Lag	%	<20%	<20%	<20%	<20%	<20%	<20%	<20%
Max Time to build up rated voltage at Rated RPM	Sec	< 2 sec provided engine reach the rated speed						

Multi-Core Power Systems available with HT alternator

** For operation of outgoing breaker higher version of Synchronization controller is required

*Product specifications & dimensions are subject to change without notice.

GENSET SPECIFICATIONS

1000 - 6600 kVA

		OPTIPRIME 3000 / 3300 DUAL CORE K12	OPTIPRIME 3000 QUAD CORE DV12	OPTIPRIME 4000 OCTA CORE PV6	OPTIPRIME 4000 QUAD CORE DV16	OPTIPRIME 5000 QUAD CORE DV16	OPTIPRIME 5000 QUAD CORE K12	OPTIPRIME 6000 / 6600 QUAD CORE K12
Rating at Rated RPM	kVA	3000 / 3300	3000	4000	4000	5000 kVA	5000 kVA	6000 / 6600 kVA
	kW	2400 / 2640	2400	3200	3200	4000	4000	4800 / 5280
Voltage / Frequency / Power Factor		415V/50Hz/0.8Pf	415V/50Hz/0.8Pf	415V/50Hz/0.8Pf	415V/50Hz/0.8Pf	415V/50Hz/0.8Pf	415V/50Hz/0.8Pf	415V/50Hz/0.8Pf
Noise Level		75dBA at 1 m	75dBA at 1 m	75dBA at 1 m	75dBA at 1 m	75dBA at 1 m	75dBA at 1 m	75dBA at 1 m
Configuration Options		4	4	3	4	4	4	3
Fuel		Diesel	Diesel	Diesel	Diesel	Diesel	Diesel	Diesel
Fuel Consumption*	At 100% Load	589	584	776	760	760	950	1178
	At 75% Load	437	480	608	588	588	718	874
	At 50% Load	298	340	492	426	426	504	596
Fuel Tank Capacity		External Tank	990 Quad Core	990 Quad Core	990 Quad Core	990 Quad Core	External Tank	External Tank
Tentative dimensions *Customized solution, actual may vary as per customer requirement and layout	Length	4303	7606	6800	9810	4505	7606	7606
	Width	4244	6340	2400	4560	3390	4244	4244
	Height (With Silencer)	2819	2575	5800	2800	3000	3120	3120
	Genset Alignment / Configuration	Kg Parallel	Parallel	Stacking+Parallel+Series	Parallel	Stacking	Parallel	Parallel
Dry weight of Genset with Canopy (Approx.)		37500	37000	37500	52000	26752	74000	74000

^ Tolerances Apply: # With 0.845 Specific Gravity of diesel (5% Tolerance) || \$ These weight are for handling & transportation only, ±5% tolerance apply || * Efficiency of Alternator as per standards IEC 60034-1 || ** For operation of outgoing breaker higher version of Synchronization controller is required || For Site Conditions other than standard operating conditions consult Kirloskar Oil Engines Ltd. || For Site specific layout consult Kirloskar Oil Engines Ltd to ammend the Genset alignment/configuration

*Product specifications & dimensions are subject to change without notice.

ENGINE SPECIFICATIONS

		OPTIPRIME 3000 / 3300	OPTIPRIME 3000 / 3300	OPTIPRIME 4000	OPTIPRIME 4000	OPTIPRIME 5000	OPTIPRIME 5000	OPTIPRIME 6000 / 6600
Engine Model		12K4300-E2	DV12ETA 4G2	D15L6	DV16ETA 4G3	DV16ETA G3	12K4300-E1	12K4300-E2
Rated output (Prime continuous ratings as per ISO 8528-1)	kW	1295 Dual Core	662 Quad Core	447 Octa Core	889 Quad Core	889 Quad Core	1097 Quad Core	1295 Quad Core
	HP	1760 Dual Core	900.6 Quad Core	608 Octa Core	1210 Quad Core	1210 Quad Core	1492 Quad Core	1760 Quad Core
Number of Cylinders	No.	12 Dual Core	12 Quad Core	6 Octa Core	16 Quad Core	16 Quad Core	12 Quad Core	12 Quad Core
Cubic Capacity	Ltrs	51.72 Dual Core	23.88 Quad Core	14.76 Octa Core	31.84 Quad Core	31.84 Quad Core	51.72 Quad Core	51.72 Quad Core
Lube Oil Change Period	hrs.	500	500	500	500	500	500	500
Lube Oil Sump Capacity	Ltrs	285 Dual Core	73 Quad Core	42 Octa Core	130 Quad Core	130 Quad Core	285 Quad Core	285 Quad Core
Coolant Capacity	Ltrs		173.9 Quad Core	250 Octa Core	180 Quad Core	180 Quad Core	355 Quad Core	
AdBlue / DEF Capacity	Ltrs	NA	45 Quad Core	45 Octa Core	NA	NA	NA	NA

ALTERNATOR SPECIFICATIONS

Insulation Class		Class H	Class H	Class H	Class H	Class H	Class H	Class H
Max Voltage Dip at Full Load 0.8 pf Lag	%	<20%	<20%	<20%	<20%	<20%	<20%	<20%
Max Time to build up rated voltage at Rated RPM	Sec	< 2 sec provided engine reach the rated speed						

Multi-Core Power Systems available with HT alternator

** For operation of outgoing breaker higher version of Synchronization controller is required

*Product specifications & dimensions are subject to change without notice.

BEYOND EFFICIENCY

A Closer Look at Cutting-Edge Technology

Top Mounted Radiator



Low Emission ←



500 Hours Lube Oil
Change Period ←



→ Efficient CRDi System



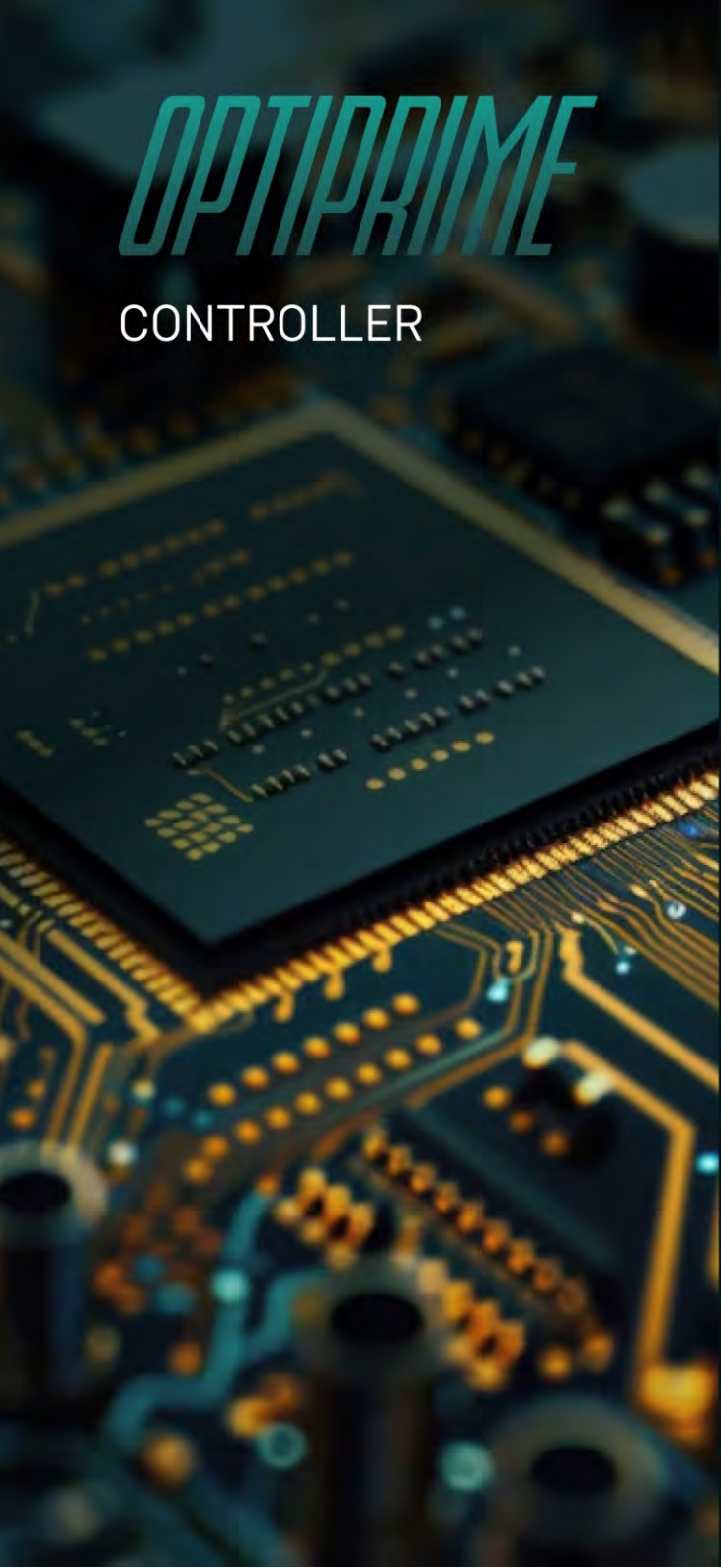
→ Compact, Robust
and Efficient



Microprocessor Based Controller

Graphical LCD Display

Advanced Monitoring and Diagnostics



OPTIPRIME

CONTROLLER

THE BRAIN BEHIND OUR POWER SYSTEMS

Multi-Core operation for superior performance and maximizing efficiency

HMI and SCADA Integration

Provides a user - friendly interface for local and remote monitoring and control, as well as seamless integration with BMS and DMS systems

CAN-based

Communication Utilizes a robust CAN bus network for seamless communication between the controller and generator components

Fast Synchronization

Enables rapid synchronization of incoming gensets to the bus, minimizing downtime and disturbances

Adaptive Control

Utilizes AI-based optimization and predictive control to adapt to varying load conditions and optimize efficiency

Fault Tolerance

Incorporates redundancy measures and fail-safe mechanisms to ensure high availability and prevent single points of failure

Efficient Load Sharing

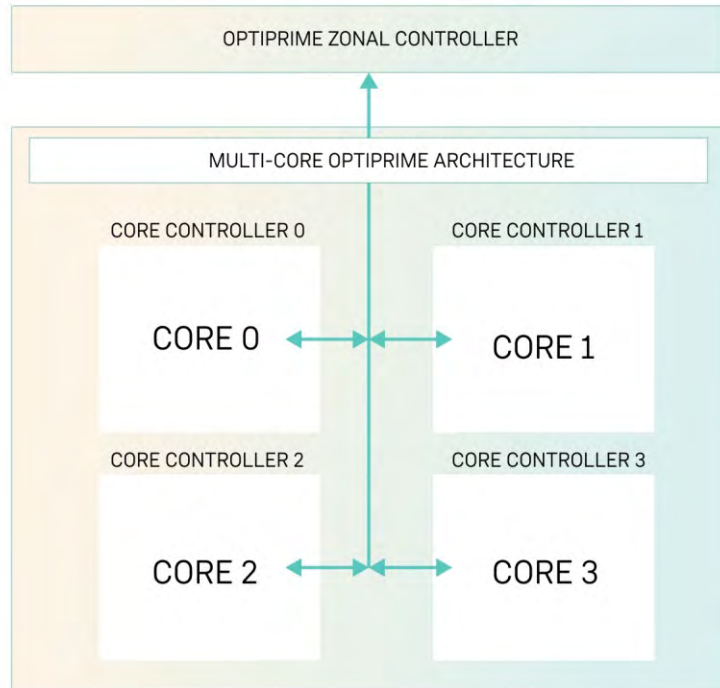
Employs advanced algorithms for precise load sharing between gensets, ensuring optimal fuel consumption and performance

Zonal Centralized Control

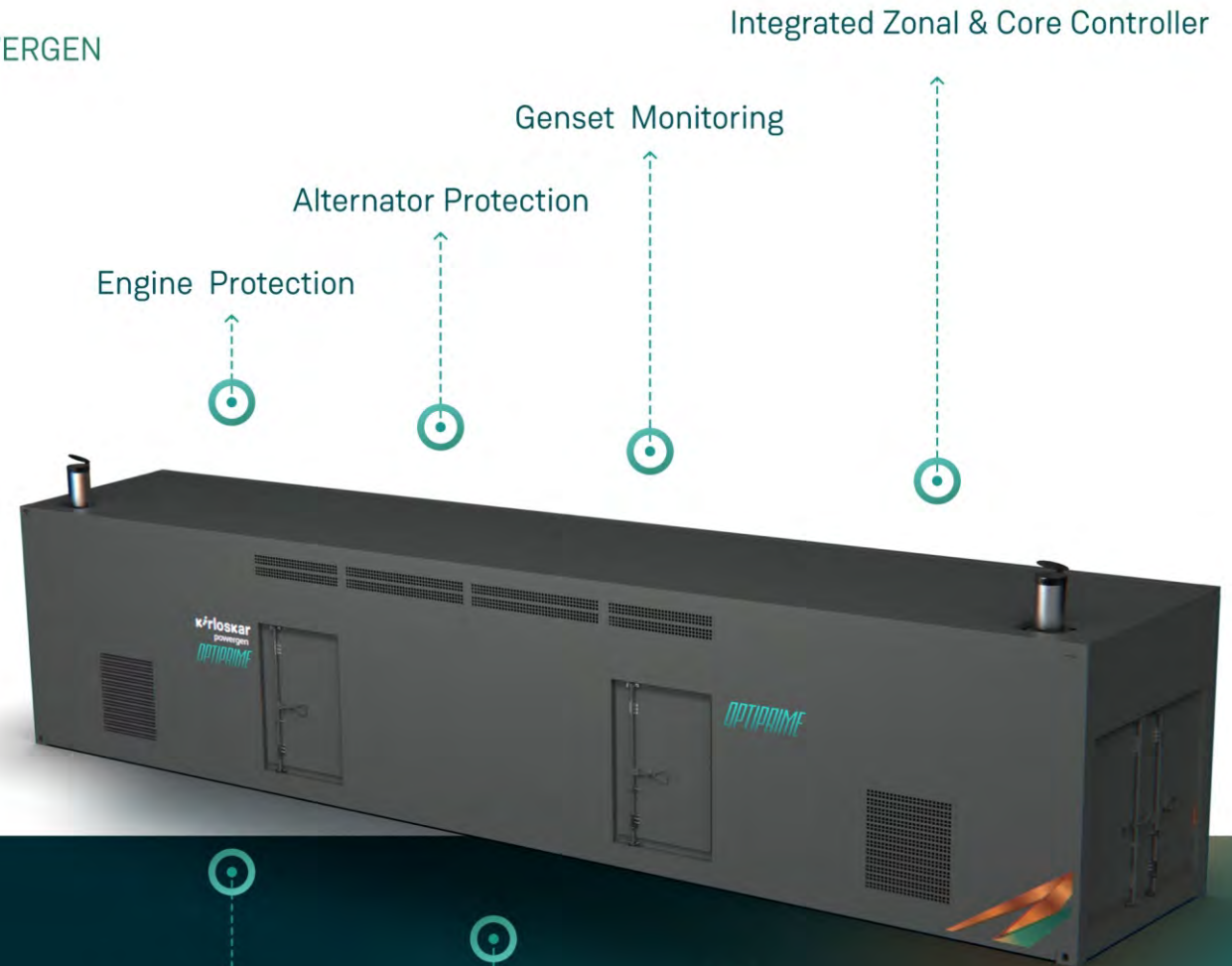
Engineered design for domain centralized control to ensure seamless transition between zonal and Core control system ensuring inherent redundancy

MULTI-CORE OPERATIONS

TAILOR MADE SOLUTION DRIVEN BY KIRLOSKAR POWERGEN



- Highly reliable OPTIPRIME Controller handles multi core operations driven by individual core controllers.
- Controller handles multiple control loops (Speed, Voltage, Phase Sync) driving superior performance.



Breaker Monitoring

In-Built Load Sharing Module

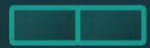
Run-Up Synchronization

In-Built Neutral Interlocking Logic

MULTI-CORE : INNOVATION

COMBINED WITH OPTIMAL EFFICIENCY

Multiple configuration options available
based on customer and site requirements



Series



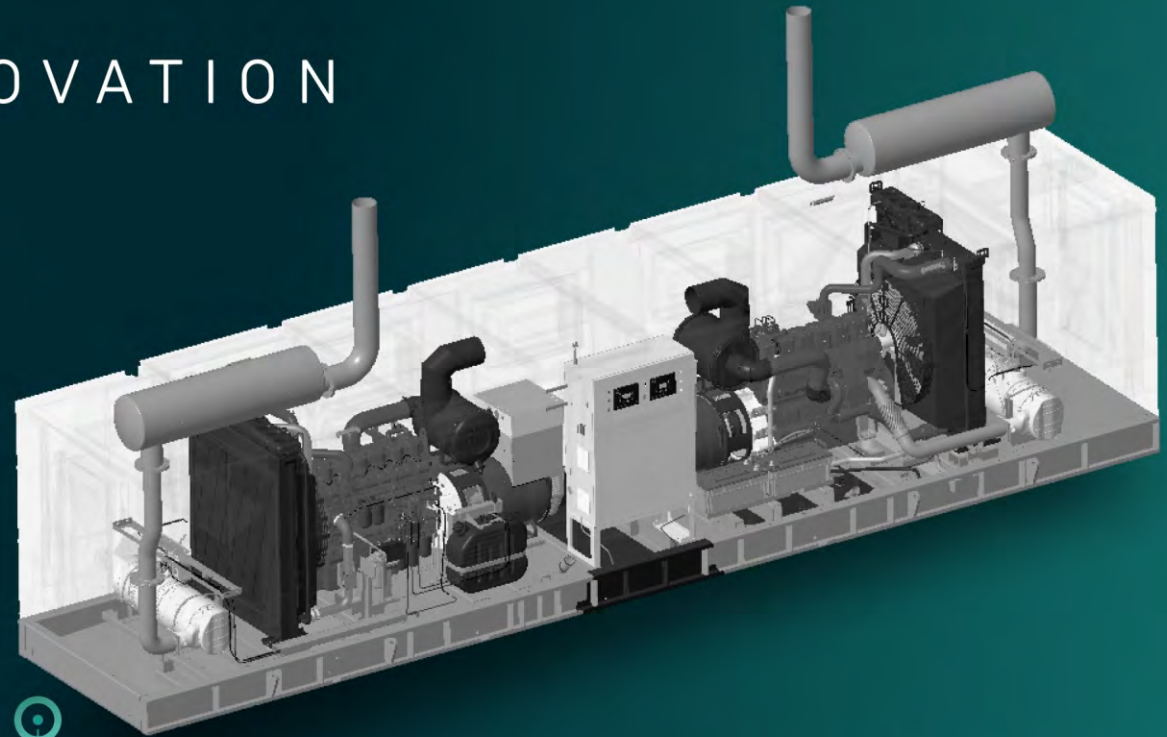
Parallel



Stack

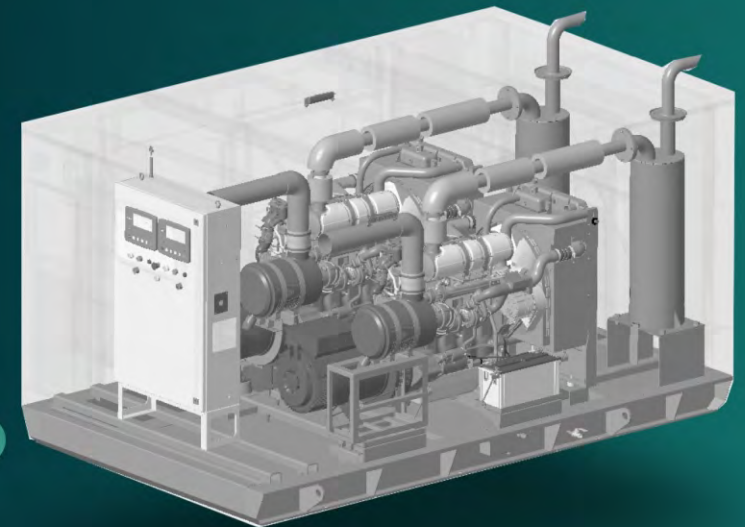


L - Shape



SERIES

PARALLEL



PROJECT SOLUTION

1000 kVA - 6600 kVA

Kirloskar Powergen's Project Solutions offers comprehensive suite of services, encompassing end-to-end engineering solutions tailored to meet diverse client requirements.

This includes product customization, site-specific adaptations, intricate design and modification capabilities, seamless system integration and robust MEP solutions. Driven by a commitment to delivering turnkey solutions, Kirloskar Powergen empowers customers to personalize their products, ensuring alignment with their unique operational needs and project specifications.

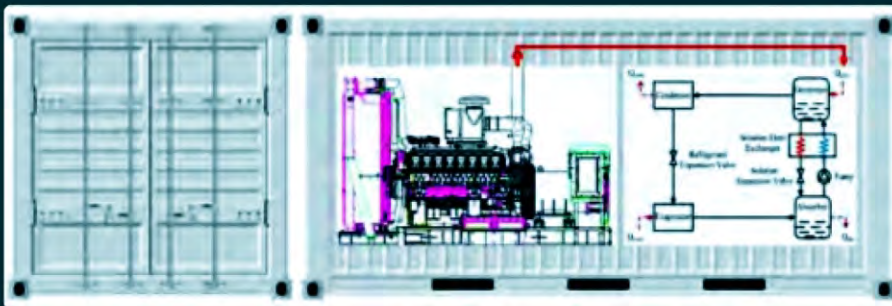


OPTIPRIME Trigeneration Series

Kirloskar's OPTIPRIME Trigeneration Series delivers a revolutionary solution for businesses with critical HVAC loads, offering uninterrupted power and cooling while significantly reducing diesel consumption. By integrating a trigeneration cycle within the genset, OPTIPRIME allows for downsizing your DG, providing rapid startup times and seamless Automatic Load Transfer / Auto Mains Failure (AMF) to ensure instant power and cooling during outages.

The innovative waste heat recovery system powers grid-independent absorption cooling, extending UPS runtime and optimizing chiller performance, all while maintaining stable and efficient HVAC support for sensitive systems like server rooms.

This prime-rated, standby-ready solution activates in seconds, providing both power and cooling, eliminating thermal build up and maximizing operational uptime, making it the ideal choice for businesses demanding reliability and efficiency.



SUSTAINABILITY

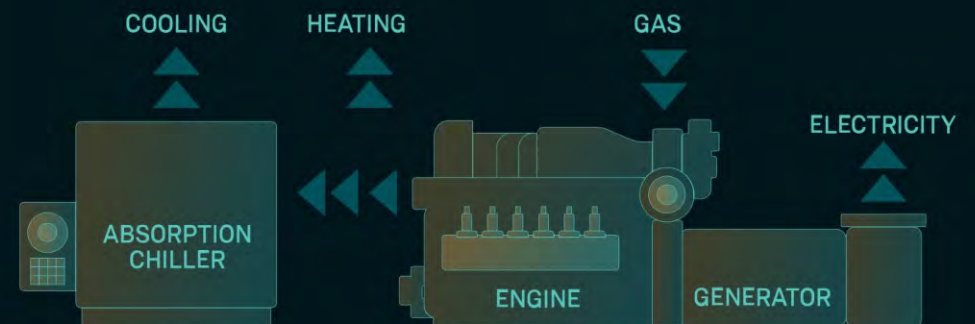


ECONOMIC ADVANTAGES

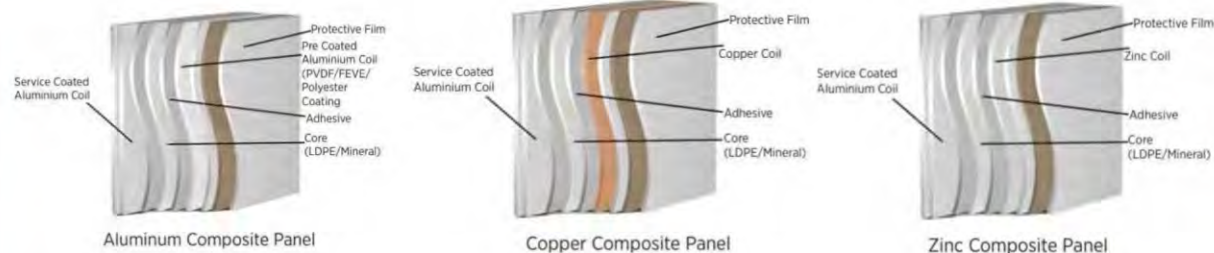
- 30-40% reduction in fuel costs
- Up to 50% savings on electricity costs for cooling

ENVIRONMENTAL BENEFITS

- 30-50% reduction in CO2 emissions
- Supports ESG & Green Building compliance
- A step towards Net Zero energy goals
- Payback within 1.5 to 4 years



THE MOST ADVANCED MATERIALS FOR ACOUSTICS & HAZARD MANAGEMENT



UNMATCHED NOISE ATTENUATION

Our composite material offers clear advantages over traditional metal sheets by reducing noise through the combination of materials with different densities and textures.

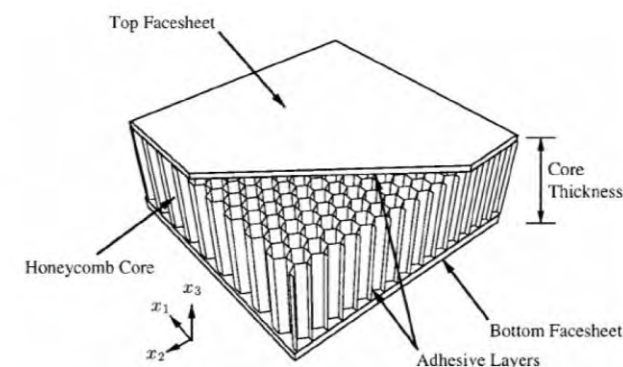
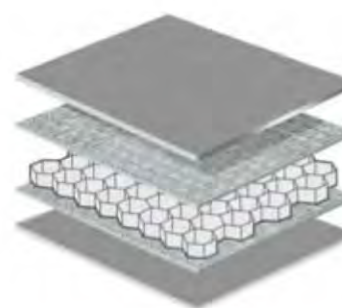
Our panels absorb and dissipate sound energy across a wide frequency range, effectively trapping sound waves and preventing them from travelling further.

This noise reduction is achieved through layered materials within the panels that act as both sound barriers and absorbers.

HONEYCOMB PANEL FOR UNMATCHED FIRE SAFETY

The unmatched strength of honeycomb panels, particularly aluminium honeycomb panels is a defining characteristic that sets them apart as a superior choice for a wide array of installations and applications. Their unique hexagonal core significantly delays fire spread, maximizing evacuation time, and they are inherently non-combustible. This combination of strength and fire resistance, coupled with their lightweight nature, is crucial in aerospace, metro coaches, and theaters.

We utilize this technology to provide our customers with custom fire-proof power solutions, ensuring their safety and minimizing risk.



NEXT-GEN POWER FOR DATA CENTRES

Data centres are the backbone of the digital world, demanding uninterrupted power, high efficiency, and seamless integration. Our Optiprime Series—an advanced multicore power generation system designed for superior performance, unmatched efficiency, and seamless integration with data centre infrastructure.

Scalability & Load Flexibility

Engines turn on/off based on demand, ensuring efficiency.



Lower Cost & Compact Footprint

Factory-fitted synchronization, lowest cost of ownership.



Fuel Efficient & Redundant

Optimized load sharing, fail-safe design.



Industry-Leading Compact Design

Maximized power density, smallest footprint.



Global Emission Compliance

Meets CPCB IV+, EURO V, and Tier IV Final standards.



Alternate Fuels Ready

Available in Natural Gas, Hydrogen, and other alternatives.



POWERING YOUR DATA CENTRE NEEDS

OPTIPRIME MODEL	KEY FEATURE
640 kVA Dual Core	Ideal for data centres with footprint constraints
1000 kVA Dual Core	A marvel of engineering for high reliability
2500 kVA Dual Core	Designed from the ground up for large-scale data centres
6600 kVA Quad Core	The ultimate solution for high-density power demands



FUTURE PROOFING YOUR DATA CENTRES

Waste Heat Recovery & Trigenation –
Convert waste heat into power & cooling.

AI-driven Remote Monitoring &
Predictive Maintenance.

Seamless Integration with
BMS, UPS & switchgear.

Up to 50% reduction in CO₂ & NO_x
emissions.

Supports ESG compliance &
Net Zero targets.

Maximized efficiency up to
85% with Trigenation.

PERFECT FOR LARGE-SCALE DATA CENTERS

DATA CENTRE LOAD	OPTIPRIME MODEL	COOLING CAPACITY (TR)
Data Centre Load	Optiprime 2500	600-600 TR
1-5 MW	Optiprime 5000	800-1000 TR
5-10 MW	Optiprime 6600	1000-2000 TR

6600 KVA - INDUSTRY FIRST QUAD CORE SYSTEM

BEST-IN-CLASS NOISE ATTENUATION & FIRE SAFETY

GLOBAL EMISSION COMPLIANCE - CPCB IV+, EURO V, TIER IV FINAL

ENERGY MANAGEMENT SYSTEM

Multi-Core patented hybrid technology

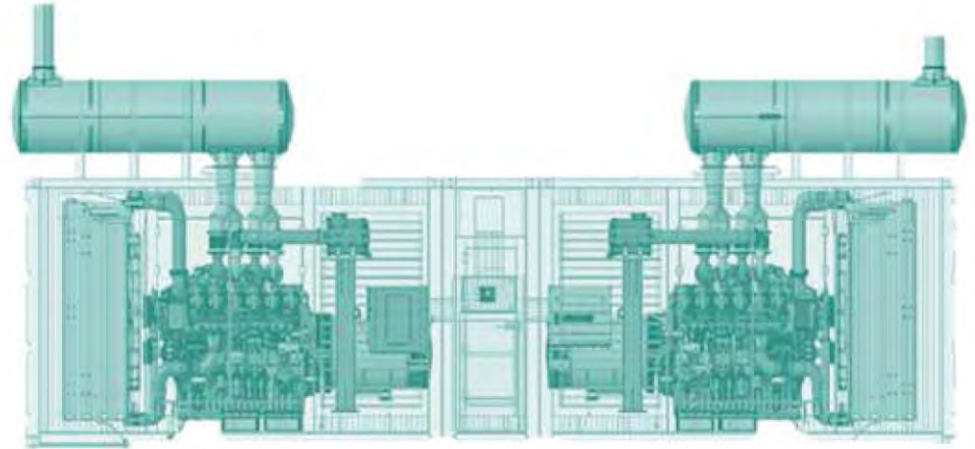
OPTIPRIME offers state of the art control system which is compatible with large Projects, Building Management Systems, SCADA and can be integrated seamlessly with Battery Storage Systems



Advanced OPTIPRIME Control System -> Integrates with your DCMS/BMS/EMS



Integrated Battery Energy Storage System -> Custom built to your requirement



OPTIPRIME: The Complete turnkey solution

- Customized engineered solution
- Complete aftermarket service support with multiple service solution contract options available
- Installation & Commissioning
- Exhaust stack & Sync panel



Reduction in Total Cost of Ownership

- Backed by Kirloskar warranty
- Improved efficiency at partial loads
- Better fuel consumption
- Leverage indigenized parts



Increased Flexibility through Hybrid ESS + ICE

- Power redundancy: N+1, 2N & 2N+1
- Compact footprint
- Enhance Reliability
- Ease of handling and transportation
- Fast Paralleling

OPTIPRIME

WORLD'S SMALLEST
1000 kVA GENSET

LOWEST FOOTPRINT



Length : 3400 mm
Width : 2400 mm
Height : 2900 mm

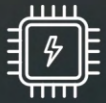
kirloskar
powergen

ADVANCED MANUFACTURED GENSETS



BENEFITS

- World's most compact footprint
- Easy to handle
- Modular Scalability for Future Growth
- Robust Redundancy for Uninterrupted Performance
- Engineered for Better sound attenuation
- Seamless Integration with Building Management Systems
- Advanced Remote Monitoring
- Visually Compelling Enclosure



MICROGRID SOLUTION



WHAT IS A MICROGRID?

Optimizing Power and Managing Loads Efficiently

A Microgrid is a self-sufficient energy system that integrates multiple power sources, optimizes energy distribution, and ensures seamless operation. Designed for reliability, resilience, and sustainability, it facilitates decentralized energy management, allowing industries and businesses to optimize costs and minimize environmental impact.

Key Features



Smart Local Energy Network

Connects power sources and loads within a unified system for efficient energy management.



Multi-Source Integration

Combines renewable energy, conventional power, and storage systems for continuous supply.



Advanced Load Management

Supports industrial & commercial applications with balanced power distribution.



Intelligent Controllers

Ensures stable power flow, seamless grid synchronization, and resilience against disruptions.



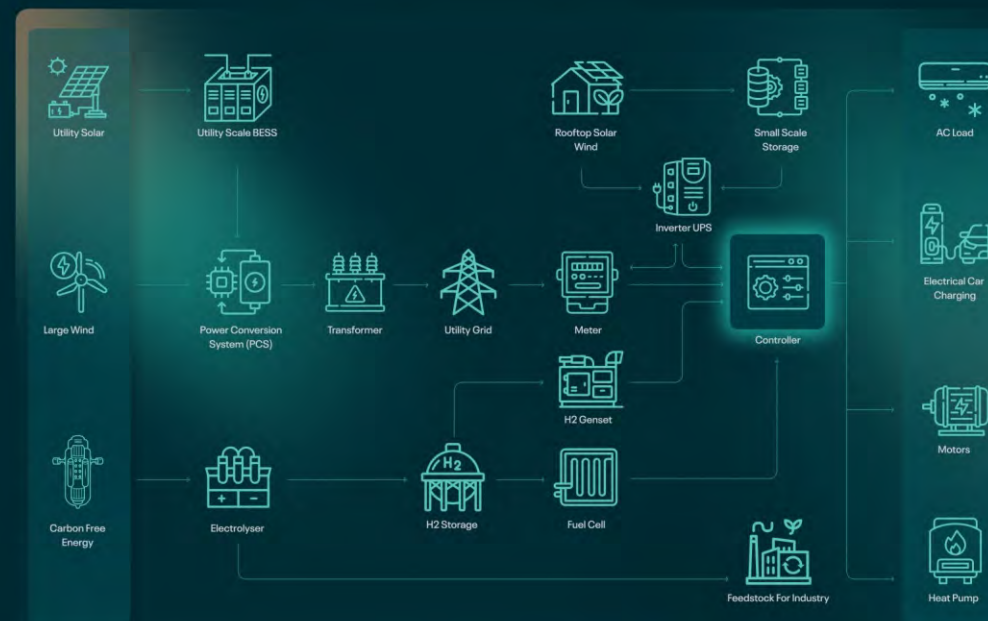
THE ENERGY TRANSFORMATION

The energy landscape is rapidly transforming with the growing adoption of renewable energy, digitalization, and the shift away from centralized power grids. Our latest state-of-the-art **MICROGRID** solution is the perfect response to the rising demand for renewable energy and cleaner power. Our Microgrid solutions, which can operate independently or alongside the main grid, provide flexible and sustainable energy for diverse needs, ranging from remote areas to industrial facilities.

Microgrid integrates energy sources like solar, wind, and biomass with energy storage and intelligent control systems, optimizing power use and reducing reliance on fossil fuels. Kirloskar Powergen leads this revolution by offering complete solutions, including power generators, power storage, and advanced controllers, to help businesses and communities build sustainable Microgrid solutions.

Microgrid systems are revolutionizing energy by creating localized, resilient power systems. These systems combine renewable sources and advanced control technologies, ensuring reliability and reducing transmission losses. They help optimize energy use, even during grid disruptions, and improve overall efficiency.

Our solution plays a crucial role in achieving decarbonisation and sustainability, and the integration of Small Modular Reactors (SMRs) further strengthens its impact. SMRs provide a stable, low-carbon energy source that complements renewable energy systems and enhances energy independence. SMRs not only improve grid reliability by offering a consistent power supply but also contribute significantly to global decarbonisation efforts, supporting the broader transition to clean energy.



BENEFITS

Reliable Power, Sustainable Solutions

Our Microgrid solution offers a range of benefits - Leading the Energy Transition



Unmatched Resilience & Reliability

Ensures uninterrupted power supply, even during grid failures, safeguarding mission-critical operations for industries, hospitals, data centres, and remote locations.



Integration of Renewable Energy

Efficiently integrates solar, wind, and other renewable sources, reducing dependence on fossil fuels while maximizing energy efficiency through advanced smart controls.



Significant Cost Savings

Our solution optimizes energy usage, reducing grid dependency and operational costs. With real-time monitoring and demand-side management, businesses can avoid peak demand charges and maximize efficiency.



Environmental Sustainability

By leveraging renewable energy and reducing carbon footprints, our microgrid solutions support corporate sustainability initiatives, ensuring compliance with global environmental standards.



Scalability & Grid Compatibility

Designed for flexibility, our solutions seamlessly integrate with existing infrastructure, offering scalability for future expansions without disrupting operations.

KIRLOSKAR'S SOLUTION

We collaborate closely with customers to envision and shape microgrid projects. Our process starts with a comprehensive Microgrid Energy Master Plan, meticulously analysing various scenarios to create a multi-year roadmap of capital and operational investments. By delivering a robust business case underpinned by investment-grade engineering, we provide a solid foundation for project planning and financing. We offer end-to-end solutions, from design and implementation to ongoing operations and maintenance.

YOUR INPUTS



- Load Profile
- Existing Energy Resources
- Expansion Plans
- Desired Microgrid Outcomes



OUR SOLUTIONS



LET'S WORK TOGETHER TO BUILD A
SUSTAINABLE ENERGY FUTURE!

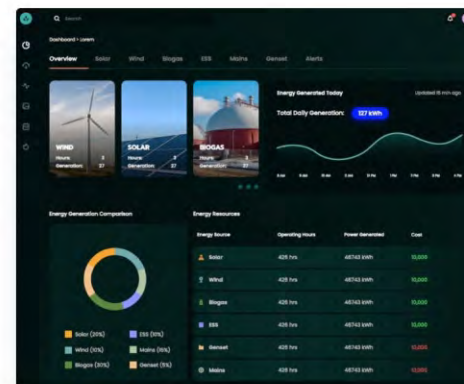
Decarbonisation Pathways for a Greener Tomorrow

Custom-Tailored Solutions designed to meet unique energy demands.

Multi source Integration ensuring compatibility with existing infrastructure.

Real-Time Monitoring with advanced analytics and alerts for optimal performance.

One-Window Solution for comprehensive service from design to maintenance.



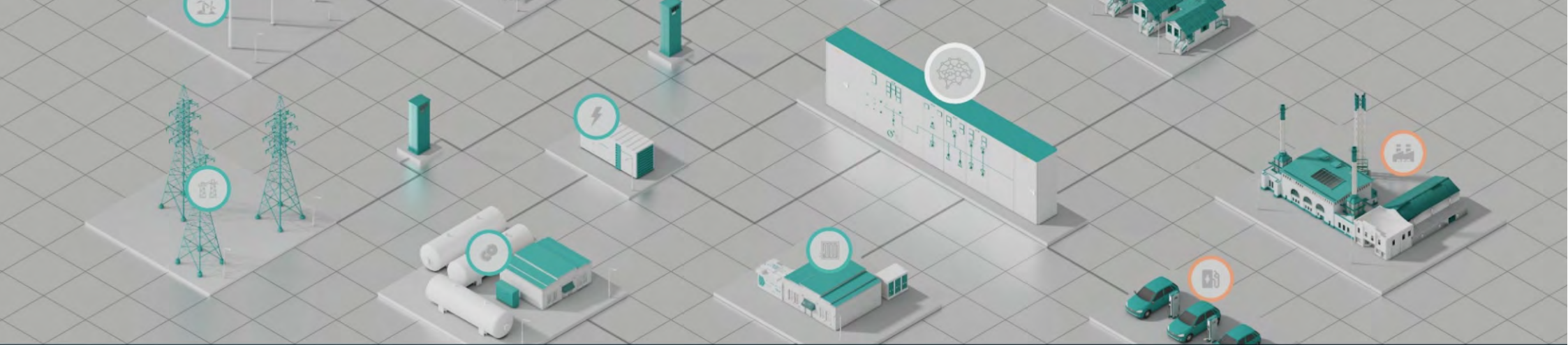
FUEL AGNOSTICS

We offer 4 de-carbonization pathways designed to empower our customers on their journey towards a sustainable, low carbon future. These tailored solutions provide a clear roadmap to reducing emissions, optimizing energy usage, and embracing cleaner technologies, ensuring that your de-carbonization goals are met efficiently and effectively.



ELECTROLYZERS & FUEL CELLS





KIRLOSKAR KAGAL PLANT

MICROGRID

Anchored By Kirloskar Powergen

Location: Kirloskar Manufacturing Plant.
Kagal, Maharashtra
Solution: Microgrid

MICROGRID SOLUTION

Power Source

- Installed base (Solar) 40 kWp
- Installed base (Wind) 36 kWp
- Mains Rating 40 kWp
- Battery Storage Rating 40 kWh
- Power generator Rating 25 kVA

Load

- Critical Load 24 kWp
- Non Critical Load 6 kWp



KIRLOSKAR KAGAL PLANT

CANTEEN MICROGRID GENERATION AND CONSUMPTION ANALYSIS



SMART ENERGY SAVINGS WITH MICROGRID SYSTEM

Our Central Kitchen achieved a 42% Reduction in Electricity Consumption averaging 5,473 units saved per month. After installing the advanced Microgrid System at our Central Kitchen, we have achieved remarkable energy savings and efficiency.

The Microgrid Panel with Energy Storage System (E.S.S.) efficiently integrates and manages multiple energy sources for the Central Kitchen Building, including:

- 40 kW Solar Power
- 30 kW Wind Energy
- Diesel Generator (DG) Set
- State Electricity Board Supply

Power Consumption Before & After Microgrid Installation

Before:

13,319 kWh

(Avg. over 3 months)

After:

7,846 kWh

(Avg. over 3 months)

INTEGRATED ENERGY SOURCES OF OUR MICROGRID



ENERGY STORAGE SYSTEM

The energy storage system captures surplus energy, storing it when generation exceeds demand and releasing it during peak consumption periods. Our advanced battery storage system has 40 kWh capacity



On-Site Renewable Energy – Solar

Harnessing the power of the sun, our system utilizes photovoltaic (PV) panels to generate clean and renewable energy. This eco-friendly approach supports sustainability and has an installed capacity of 40 kWp.



On-Site Renewable Energy – Wind

Wind energy is efficiently harnessed through strategically placed wind turbines, contributing to a diversified and resilient power mix. Our system is designed to optimize energy capture, and has installed capacity of 36 kWp.



Power Generator

Our microgrid includes a diesel generator rated at 25 kVA, serving as a backup power source. It ensures uninterrupted electricity supply during outages or peak demand periods. The system is also compatible with various energy sources, including biogas, methanol, ethanol, hydrogen, and petrol.



MAINS

To ensure seamless operation, our microgrid is connected to the larger electricity grid, serving as an essential backup power source. Our system features a mains connection with a capacity of 40 kWe, ensuring continuous and stable energy access.

KIRLOSKAR POWERGEN GENSETS

LHP | MHP | HHP

**7.5 kVA to
2000 kVA**



CPCB IV+

GENSETS

(7.5 kVA - 750 kVA)

EXHAUST GAS TREATMENT (Up to 750 kVA)

- DOC and SCR system sets off the reaction to meet the latest Emission Norms
- EGR system to reduce the level of Nox and HC emission **(10 kVA to 58.5 kVA)**
- Drastic reduction in Particle Matter

SUPPLY MODULE AND DCU (82.5 kVA & above)

- Control and Monitor the DEF

DEF TANK (82.5 kVA & above)

- DEF / Aqueous urea to set off chemical reaction with Exhaust Gas
- Tank size is optimized in accordance to DEF consumption

CANOPY

- Ease of access and serviceability
- Aesthetically Designed
- Insulation conforms to UL94-HF1 Class for flammability
- Sound and Weather Resistant Enclosure

ENGINE

- Efficient CRDi System02E Series:
- Low emission, high efficiency engine
- Compact Robust and Rugged Design
- 500 hours lube-oil change period

CONTROLLER

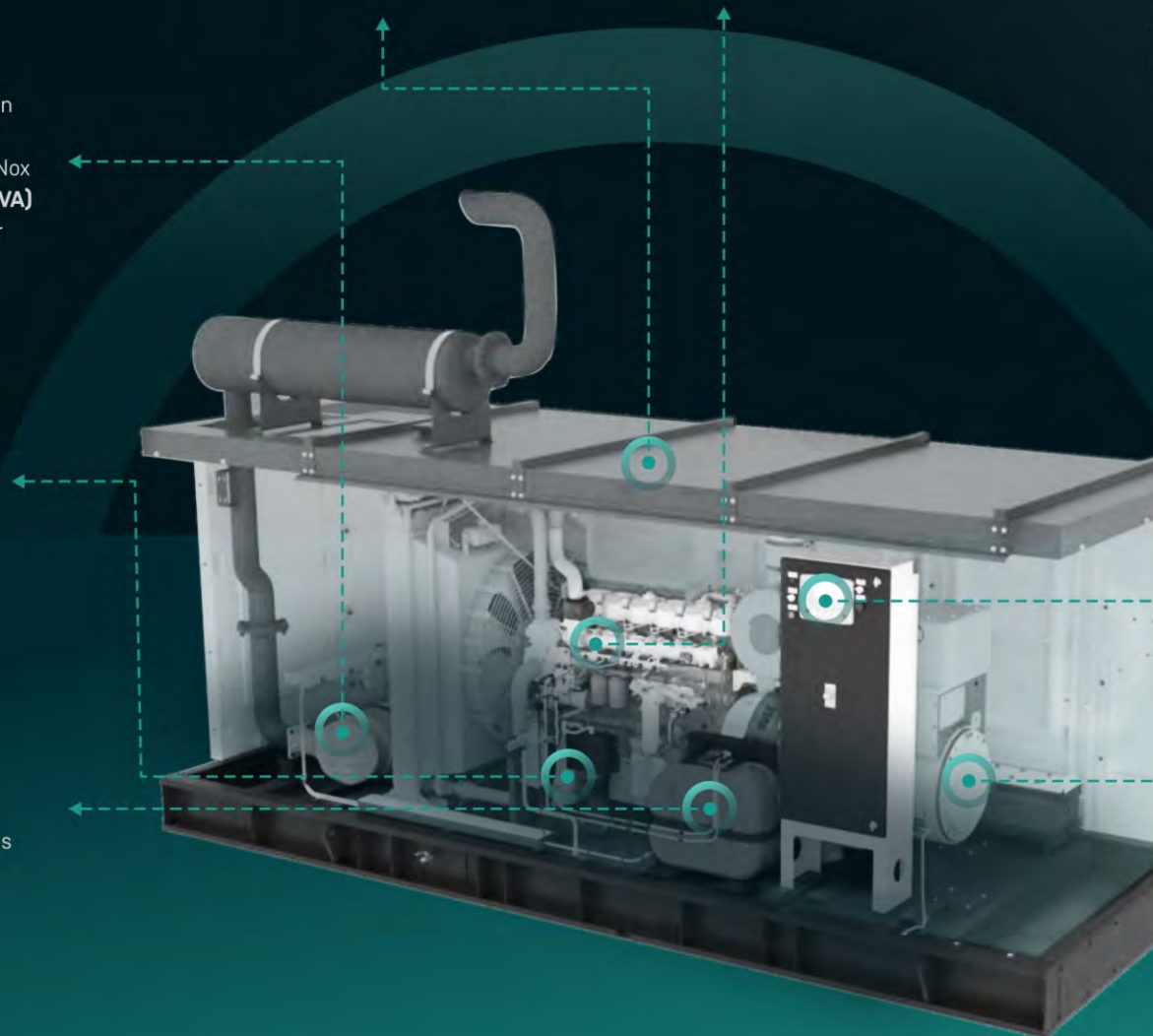
- Microprocessor Based
- Advanced Monitoring and Diagnostic Features
- Graphical LCD Display **(250 kVA & above)**
- Integrable with AMF, Synchronization and communication compatible. **(250 kVA & above)**

ALTERNATOR

- Special Windings to reduce harmonics
- Vacuum pressure impregnation and epoxy gel coating

IN-BUILT SILENCER (Up to 25 kVA)

- Lower Noise Levels
- Space Savings



TECHNICAL SPECIFICATIONS LHP

Prime Rating at rated rpm (as per ISO 8528)	kVA	7.5	10	15	20	25	30	40
	kW	6	8	12	16	20	24	32
Genset Model		KG4-7.5SAS1	KG4-10WS1	KG4-15WS1	KG4-20WS1	KG4-25WS1	KG4-30WS1	KG4-40WS1
Frequency	Hz	50	50	50	50	50	50	50
Power factor	Lagging	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Voltage	V	230 (1φ) & 415 (3φ)				415 (3φ)		
Governing class (As per ISO 8528-5)	-	G2	G2	G2	G2	G2	G2	G2
Noise level	dBA	<75	<75	<75	<75	<75	<75	<75
Fuel tank capacity (Standard DG set)	Ltrs.	50	32	32	40	50	72	100
⚙ Weight of genset with canopy (approx..)^	Dry	Kg	620	585	605	680	770	1165
	Wet (w/o fuel)	Kg	670	590	610	690	780	1180
	Length	mm	1310	1800	1850	2180	2330	2750
Overall dimensions of genset	Width	mm	810	760	760	905	950	1050
	Height	mm	1600	1050	1050	1150	1260	1495
Electrical battery starting voltage	Volts-DC	12	12	12	12	12	12	12

ENGINE

Engine Model	-	EA10NA 4G1	3R550NA 4G1	3R550TC 4G1	3R550TA 4G1	3R550ETA 4G1	3R1190ENA 4G1	3R1190ETA 4G1
Rated output (Prime Continuous rating as per ISO 8528-1)	kW	7.3	11	15.4	18.8	26.46	30.87	41.1
	HP	10	15	20.9	25.5	36	42	56
No. of cylinder	Number	1	3	3	3	3	3	3
Cubic capacity	Ltrs	0.95	1.65	1.65	1.65	1.65	3.57	3.57
Bore x Stroke	mm	102 x 116	86x94	86x94	86x94	86x94	110 x 125	110 x 125
Rated speed	RPM	1500	1500	1500	1500	1500	1500	1500
Aspiration	NA/TC/TA	NA	NA	TC	TA	TA	NA	TA
Lube Oil change period	hrs.	500	500	500	500	500	500	500
Lube oil Sump Capacity (max)	Ltrs	5	5.95	5.95	5.95	5.95	7	7
Coolant Capacity (Engine + radiator)	Ltrs	NA	3.78	4.2	4.9	4.9	10	8.3
Adblue/ DEF capacity**	Ltrs	NA	NA	NA	NA	NA	NA	NA

ALTERNATOR

Insulation Class		Class H	Class H	Class H	Class H	Class H	Class H	Class H
Alternator efficiency (at 100% load) 0.8 pf	%	78.1	80.3	85.2	88.6	87.9	88.4	87.9
Max Voltage Dip at Full Load 0.8 pf Lag	sec	<20%	<20%	<20%	< 20 %	< 20 %	< 16 %	<16%
Max Time to build up rated voltage at Rated RPM		< 2 sec, provided engine reach the rated speed						

- NOTES:
- ^ Tolerance Apply
- ⚙ These weight are for handling & transportation only.
- NA# Inbuilt fuel tank is not applicable
- ** AdBlue used should be follow Iso 22241
- *** Efficiency of Alternator as per standards IEC:60034-1

TECHNICAL SPECIFICATIONS LHP

Prime Rating at rated rpm (as per ISO 8528)	kVA	58.5	82.5	100	125	160	200	200HD	250
	kW	46.8	66	80	100	128	160	160	200
Genset Model		KG4-58.5WS	KG4-82.5WS1	KG4-100WS1	KG4-125WS1	KG4-160WS11	KG4-200WS1	KG4-200WS1	KG4-250WS1
Frequency	Hz	50	50	50	50	50	50	50	50
Power factor	Lagging	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Voltage	V	415 (3φ)							
Governing class (As per ISO 8528-5)	-	G2	G3	G3	G3	G3	G3	G3	G3
Noise level	dBA	<75	<75	<75	<75	<75	<75	<75	<75
Fuel tank capacity (Standard DG set)	Ltrs.	165	200	230	230	400	400	460	600
⚙ Weight of genset with canopy (approx..)^	Dry Kg	1460	1800	2170	2200	2980	3080	3850	3990
	Wet (w/o fuel) Kg	1485	1860	2230	2260	3050	3150	3960	4100
Overall dimensions of genset	Length	mm	2900	3200	3200	3200	4200	4200	4550
	Width	mm	1100	1350	1350	1350	1450	1450	1700
	Height	mm	1580	1595	1796	1796	1900	1900	2005
Electrical battery starting voltage	Volts-DC	12	12	12	12	12	12	24	24

ENGINE

Engine Model	-	4R810ETA 4G1	4R1190ETA 4G1	4K1080ETA 4G1	4K1080ETA 4G1	6K1080ETA 4G1	6K1080ETA 4G2	6SL90ETA 4G1	6SL90ETA 4G2
Rated output (Prime Continuous rating as per ISO 8528-1)	kW	54.4	77.2	114.7	114.7	147.1	183.8	183.8	228
	HP	74	105	156	156	200	250	250	310
No. of cylinder	Number	4	4	4	4	6	6	6	6
Cubic capacity	Ltrs	3.24	4.76	4.32	4.32	6.48	6.48	8.86	8.86
Bore x Stroke	mm	96 x 112	110 x 125	105 x 125	105 x 125	105 x 125	105 x 125	118 x 135	118 x 135
Rated speed	RPM	1500	1500	1500	1500	1500	1500	1500	1500
Aspiration	NA/TC/TA	TA	TA	TA	TA	TA	TA	TA	TA
Lube Oil change period	hrs.	500	500	500	500	500	500	500	500
Lube oil Sump Capacity (max)	Ltrs	10	10	14	14	25	25	27	27
Coolant Capacity (Engine + radiator)	Ltrs	12.7	15.1	17.7	17.7	28.9	28.9	35.8	36.4
Adblue/ DEF capacity**	Ltrs	NA	25	25	25	25	45	45	45

ALTERNATOR

Insulation Class		Class H	Class H	Class H	Class H	Class H	Class H	Class H	Class H
Alternator efficiency (at 100% load) 0.8 pf	%	90.8	91.1	89.6	91.9	92.4	92.6	92.8	94
Max Voltage Dip at Full Load 0.8 pf Lag	sec	<20%	<20%	< 20 %	< 20 %	< 20 %	< 20 %	< 20 %	< 20 %
Max Time to build up rated voltage at Rated RPM		< 2 sec, provided engine reach the rated speed							

• NOTES:
• ^ Tolerance Apply
• ⚙ These weight are for handling & transportation only.

• NA# Inbuilt fuel tank is not applicable
• ** AdBlue used should be follow Iso 22241
• *** Efficiency of Alternator as per standards IEC:60034-1

TECHNICAL SPECIFICATIONS MHP

Prime Rating at rated rpm (as per ISO 8528)	kVA	320	320HD	400	500	500 HD	625	625 HD	750
	kW	256	256	320	400	400	500	500	600
Genset Model		KG4-320WS1	KG4-320WS(HD)	KG4-400WS11	KG4-500WS	KG4-500WS(HD)	KG4-625WS	KG4-625WS11	KG4-750WS
Frequency	Hz	50	50	50	50	50	50	50	50
Power factor	Lagging	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Voltage	V	415 (3φ)							
Governing class (As per ISO 8528-5)	-	G3	G3	G3	G3	G3	G3	G3	G3
Noise level	dBA	<75	<75	<75	<75	<75	<75	<75	<75
Fuel tank capacity (Standard DG set)	Ltrs.	600	850	850	850	990	990	990	990
Weight of genset with canopy (approx..)^	Dry	Kg	4090	6169	6220	6240	8080	8150	8760
	Wet (w/o fuel)	Kg	4200	6345	6415	6435	8300	8370	9100
Overall dimensions of genset	Length	mm	4750	5100	5575	5575	6500	6500	6800
	Width	mm	1700	2125	2125	2125	2125	2300	2300
	Height	mm	2005	2610	2610	2610	2710	2710	2715
Electrical battery starting voltage	Volts-DC	24	24	24	24	24	24	24	24

ENGINE

Engine Model	-	6SL90ETA 4G3	DV8ETA 4G1	DV8ETA 4G2	DV8ETA 4G3	DV10ETA 4G1	DV10ETA 4G2	DV12ETA 4G1	DV12ETA 4G2
Rated output (Prime Continuous rating as per ISO 8528-1)	kW	279.5	294.2	360	447.2	447.2	561.1	561.1	662
	HP	380	400	490	608	608	763	763	900.6
No. of cylinder	Number	6	8	8	8	10	10	12	12
Cubic capacity	Ltrs	8.86	15.92	15.92	15.92	19.9	19.9	23.88	23.88
Bore x Stroke	mm	118 x 135	130 x 150	130 x 150	130 x 150	130 x 150	130 x 150	130 x 150	130 x 150
Rated speed	RPM	1500	1500	1500	1500	1500	1500	1500	1500
Aspiration	NA/TC/TA	TA	TA	TA	TA	TA	TA	TA	TA
Lube Oil change period	hrs.	500	500	500	500	500	500	500	500
Lube oil Sump Capacity (max)	Ltrs	27	40	40	40	50	50	73	73
Coolant Capacity (Engine + radiator)	Ltrs	36	55	63.2	63.2	81.7	81.7	173.9	173.9
Adblue/ DEF capacity**	Ltrs	45	45 x 2	45 x 2	45 x 2	45 x 2	45 x 2	45 x 2	45 x 2

ALTERNATOR

Insulation Class		Class H	Class H	Class H	Class H	Class H	Class H	Class H	Class H
Alternator efficiency (at 100% load) 0.8 pf	%	95.3	94.3	93.4	94.8	94.6	95.7	95.7	94.7
Max Voltage Dip at Full Load 0.8 pf Lag	sec	< 20 %	<20%	<20%	<20%	<20%	< 20 %	< 20 %	< 20 %
Max Time to build up rated voltage at Rated RPM		< 2 sec, provided engine reach the rated speed							

• NOTES:
• ^ Tolerance Apply
• ⚙ These weight are for handling & transportation only.

• NA# Inbuilt fuel tank is not applicable
• ** AdBlue used should be follow Iso 22241
• *** Efficiency of Alternator as per standards IEC:60034-1

TECHNICAL SPECIFICATIONS HHP

1010 kVA to 1500 kVA | 2000 kVA (from 2026)



Prime Rating at rated rpm (as per ISO 8528)	kVA	1010	1250	1500
	kW	808	1000	1200
Genset Model		KG1-1010WS	KG1-1250WS	KG1-1500WS
Frequency	Hz	50	50	50
Power factor	Lagging	0.8	0.8	0.8
Voltage	V	415 (3φ)		
Governing class (As per ISO 8528-5)	-	G3	G3	G3
Noise level	dBA	Contact Kirloskar Oil Engines Ltd for Details		
Fuel tank capacity (Standard DG set)	Ltrs.	990	NA#	NA#
Weight of genset with canopy (approx..)^ Dry Wet (w/o fuel)	Kg	13200	18250	18750
	Kg	19800	27375	28125
	mm	7800	8500	8500
Overall dimensions of genset	mm	2300	2560	2560
	mm	2713	3120	3120
Electrical battery starting voltage	Volts-DC	24	24	24

ENGINE

Engine Model	-	DV16ETA G3	12K4300-E1	12K4300-E2
Rated output (Prime Continuous rating as per ISO 8528-1)	kW	889	1097	1295
	HP	1210	1492	1760
No. of cylinder	Number	16	12	12
Cubic capacity	Ltrs	31.84	51.73	51.73
Bore x Stroke	mm	130x150	170x190	170x190
Rated speed	RPM	1500	1500	1500
Aspiration	NA/TC/TA	TA	TA	TA
Lube Oil change period	hrs.	500	500	500
Lube oil Sump Capacity (max)	Ltrs	130	285	285
Coolant Capacity (Engine + radiator)	Ltrs	180	355	355
Adblue/ DEF capacity**	Ltrs	NA	NA	NA

ALTERNATOR

Insulation Class		Class H	Class H	Class H
Alternator efficiency (at 100% load) 0.8 pf	%	95.1	95.1	95.4
Max Voltage Dip at Full Load 0.8 pf Lag	sec	<20%	<20%	<20%
Max Time to build up rated voltage at Rated RPM		< 2 sec, provided engine reach the rated speed		

- NOTES:
- ^ Tolerance Apply
- ⚙ These weight are for handling & transportation only.

- NA# Inbuilt fuel tank is not applicable
- ** AdBlue used should be follow Iso 22241
- *** Efficiency of Alternator as per standards IEC:60034-1

KIRLOSKAR POWERGEN GLOBAL GENSETS

15 kVA - 1010 kVA | 2000 kVA (from 2026)



50 Hz

TECHNICAL SPECIFICATIONS

3 PHASE, 4 WIRE, 0.8 PF

23 kVA - 1010 kVA | 2000 kVA (from 2026)

60 Hz

			LHP																		MHP								HHP							
Model		Open	25W60		39W60		50W60-3		50W60-4		80W60		96W60		127W60		150W60		202W60		248W60		275W60		352W60		440W60		550W60		660W60		825W60		1111W60	
		SAE	25WS60		39WS60		50WS60-3		50WS60-4		80WS60		96WS60		127WS60		150WS60		202WS60		248WS60		275WS60		352WS60		440WS60		550WS60		660WS60		825WS60		1111WS60	
Voltage	L-L	V	220	380	220	380	220	380	220	380	220	380	220	380	220	380	220	380	220	380	220	380	220	380	220	380	220	380	220	380	380	380				
	L-N	V	127	220	127	220	127	220	127	220	127	220	127	220	127	220	127	220	127	220	127	220	127	220	127	220	127	220	127	220	127	220	220	220		
Standby Power Output	ESP	kVA	25.3		39		50		50		80		96		127		150		202		248		275		352		440		550		660		825		1111	
		kW	20.2		31.2		40		40		64		76.8		101.6		120		161.6		198.4		220		281.6		352		440		528		660		888.8	
Prime Power Output	PRP	kVA	23		35		45		45		72		87		115		135		184		225		250		320		400		500		600		750		1010	
		kW	18.4		28		36		36		57.6		69.6		92		108		147.2		180		200		256		320		400		480		600		808	
Fuel Consumption (at % of PRP)	50%	L/Hr	3.3		4.5		5.9		5.9		9.6		10.5		11.7		16.9		22.4		28.3		31.8		41.5		48.6		61.9		73.2		85		121	
	75%	L/Hr	4.1		5.9		8.1		8.1		12.8		14.7		17.3		23.6		31.9		39.5		44.4		56		63.3		79.5		92.2		122		165	
	100%	L/Hr	5.3		7.6		10.6		10.6		17		19.2		23.6		31		42.1		51.4		56.2		72		84.9		106.7		127		155		218	
Fuel Tank Capacity	Open	L	62		70		95		95		158		158		225		230		275		340		350		552		552		680		900		880		410	
	SAE	L	40		80		99		97		150		150		195		195		270		325		325		560		560		870		930		1350		900	
Sound Level at 7m	SAE	dB (A)	70		70		70		70		70		70		70		70		70		70		70		70		70		75		75		80		80	
Dimensions (LxWxH) (as per GA Drawing)	Open	cm	141 x 104 x 133		150 x 109 x 134		175 x 109 x 136		175 x 109 x 136		211 x 106 x 146		211 x 106 x 146		224 x 105 x 169		224 x 105 x 169		260 x 110 x 183		311 x 154 x 170		311 x 157 x 190		339 x 166 x 220		339 x 166 x 220		365 x 184 x 235		379 x 200 x 238		398 x 200 x 240		454 x 210 x 251	
	SAE	cm	198 x 97 x 123		218 x 101 x 145		262 x 111 x 157		262 x 111 x 157		297 x 116 x 166		297 x 116 x 166		327 x 116 x 193		327 x 116 x 193		372 x 155 x 211		437 x 162 x 198		436 x 165 x 200		498 x 211 x 228		498 x 211 x 228		525 x 212 x 257		566 x 211 x 257		580 x 230 x 253		687 x 235 x 250	
Dry Weight (net)	Open	kg	700		790		840		840		1100		1200		1450		1510		1910		2320		2620		4250		4550		5880		5880		8650		8650	
Approx.	SAE	kg	970		1100		1310		1310		1630		1950		2120		2180		2720		3440		3670		6000		6300		7100		7830		7850		13200	

KIRLOSKAR ENGINE (LIQUID COOLED, 4 STROKE, DIESEL ENGINE)

Model		2R1040	3R1040	3R1040TA	4R1040TA	4R1040TC	4R1040TA	4K1080TA	4K1080TA	6K1080TA	6SL1500TA	6SL8800TA	DV8	DV8	DV10	DV12	DV12ETAG12	DV16 ETA G1
Engine Output	PRP	hp	27	46	62	62	90	112	170	230	279	310	400	490	608	750	900	1210
Cycles and Configuration		2 - inline	3 - inline	3 - inline	4 - inline	4 - inline	4 - inline	4 - inline	4 - inline	6 - inline	6 - inline	6 - inline	8 - V	8 - V	10 - V	12 - V	12 - V	16 - V
Bore x Stoke	mm	105 x 120	105 x 120	105 x 120	105 x 120	105 x 120	105 x 120	105 x 125	105 x 125	105 x 125	118 x 135	118 x 135	130 x 150	130 x 150	130 x 150	130 x 150	130 x 150	135 x 150
Compression Ratio		18:1	18:1	18:1	18:1	18:1	18:1	15.5:1	15.5:1	15.5:1	17.5:1	17.5:1	16.5:1	16.5:1	16.5:1	16.5:1	16.5:1	16.5:1
Displacement	L	2.08	3.12	3.12	3.12	4.16	4.16	4.32	4.32	6.48	8.86	8.86	15.91	15.91	19.9	23.88	23.88	31.86
Aspiration		Natural	Natural	TA	Natural	TC	TA	TA	TA	TA	TA	TA	TA	TA	TA	TA	TA	TA
Starting System (DC)	V	12	12	12	12	12	12	12	12	12	24	24	24	24	24	24	24	24
Governor		Mechanical	Mechanical	Mechanical	Mechanical	Mechanical	Mechanical	Electronic	Electronic	Electronic	Electronic	Electronic	Electronic	Electronic	Electronic	Electronic	Electronic	Electronic
Governor Class as per ISO8528-5		G2	G2	G2	G2	G2	G2	G2	G2	G2	G2	G2	G2	G2	G2	G2	G2	G2
Lub Oil Sump Capacity - Refill	L	5.5	7.5	7.5	7.5	9.5	9.5	14	14	18	21	21	38	38	42	42	50	125
Coolant Capacity	L	10	12	12	12	24	24	54	54	54	43	43	123	123	133	144	166	180

ALTERNATOR (BRUSHLESS, SELF EXCITED, 4 POLE ALTERNATOR)

Make	Stamford	Stamford	Stamford	Stamford	Stamford	Stamford	Stamford	Stamford	Stamford	Stamford	Stamford	Stamford	Stamford	Stamford	Stamford	Stamford	Leroy Somer	Leroy Somer
Model	SOL2MI	SIL2JI	SIL2NI	SIL2NI	UCI 224F1	UCI 224G1	UCI 274C1	UCI 274D1	UCI 274F1	UCI 274H1	UCI 274 J1 / K1	HCI 444E1	HCI 444F1	HCI 544D1	HCI 544E1	LSA49.1M75	LSA 49.3 L110	
No. of Bearings	Single	Single	Single	Single	Single	Single	Single	Single	Single	Single	Single	Single	Single	Single	Single	Single	Single	Single
Insulation	Class H	Class H	Class H	Class H	Class H	Class H	Class H	Class H	Class H	Class H	Class H	Class H	Class H	Class H	Class H	Class H	Class H	Class H
Temp. Rise at Ambient 40°C	°C	125	125	125	125	125	125	125	125	125	125	125	125	125	125	125	125	125
Wires	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
Ingress Protection	IP23	IP23	IP23	IP23	IP23	IP23	IP23	IP23	IP23	IP23	IP23	IP23	IP23	IP23	IP23	IP23	IP23	IP23
AVR Model	AS540	AS540	AS540	AS540	AS440	AS440	AS440	AS440	AS440	AS440	AS440	AS440	AS440	AS440	AS440	AS440	D350	D350
Voltage Regulation	%	+/-1%	+/-1%	+/-1%	+/-1%	+/-1%	+/-1%	+/-1%	+/-1%	+/-1%	+/-1%	+/-1%	+/-1%	+/-1%	+/-1%	+/-1%	+/-1%	+/-1%

DI - Auto start digital input is available as standard feature · SD- Shutdown · SAE : Sound Attenuated Enclosure · ESP : Emergency Standby Power rating · PRP : Prime Rated Power rating · CT : Current Transformer · LL: Line to Line · LN : Phase to Neutral · TC : Turbocharged · TA: Turbocharged Aftercooled · AVR : Automatic Voltage Regulator · MCB: Miniature Circuit Breaker · MCCB : Moulded Case Circuit Breaker

The declared fuel consumption is calculated values with 0.845 specific gravity of diesel as per BS 2869 class A2 and confirms to ISO3046 and has production tolerance of +5%.

The declared weight and dimensions are approximate and for transportation purpose. Actual weight and dimensions are may vary as per the agreed scope. Mentioned height is without tail pipe.

Definitions of Ratings as per ISO 8528-1 - PRP - PRP ratings are applicable for supplying continuous electrical power at variable load in lieu of commercial purchased power. There is no limitation to the annual hours of operation and this model can supply 10% overload for 1 hour in 12 hours. ESP - ESP rating is defines as the maximum power available during a variable electrical power sequence, under the stated operating conditions, for which a generating set is capable of delivering in the event of utility power outage. This rating is applicable for 200hours of operation per year with the maintenance intervals and procedures being carried out.

The permissible average power output over 24 hours of operation shall not exceed 70% of the ESP rating. No overload is permitted above ESP rating.

For site conditions other than the standard reference conditions of ISO8528-1 the derating is applicable. Please contact nearest KOEL office for the available power calculation at given site conditions.

The alternator models and controller models are of standard product and may vary as per the agreed scope.

Alternator may vary basis on requirement.

CUSTOMER TESTIMONIALS

REAL ESTATE

Kirloskar Powergen's 1000 kVA OPTIPRIME system delivered exceptional reliability and fuel efficiency. Their seamless process, from expert guidance to robust after sales support, was impressive. OPTIPRIME technology has significantly improved our operations and cost savings. We highly recommend Kirloskar Powergen.

HOTEL BUSINESS

We needed a dependable 400 kVA backup, and Kirloskar Powergen was able to cater our needs. Honestly it just works. Fuel efficiency is good, and their team walked us through all the technicalities and provided sales & service support. When you have a power need, you want someone who gets it done right. Kirloskar Powergen did.

HEALTH CARE

In a hospital setting, power reliability isn't a luxury, it's a necessity. We chose Kirloskar Powergen's 500 kVA OPTIPRIME system because we needed a brand we could trust and rely on for dependable backup power. When the power goes out, it kicks in instantly. That's what matters in healthcare.

Anubandhan IV+

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CPCB IV+ Diesel Genset



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Complete coverage for
5 years or 6000 hours[#]



Complete Range of Gensets

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for gensets from
7.5 to 750 kVA



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Considerable savings in
maintenance cost over
the period of 5 years



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support for prompt
resolution of breakdowns
and major issues



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at pre-commissioning &
post-commissioning of genset



Customised Package

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per genset running hours



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spares for your DG set



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by Trained and
Certified engineers.



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A Kirloskar Group Company

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POWER
FOR A

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